On the Investment Network and Development *

Lucía Casal † Julieta Caunedo‡

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Abstract

Capital accumulation, productivity growth and sectoral reallocation are salient features of economic development. These features are interconnected through the means for production and heterogeneous uses of different capital types across sectors, i.e. the investment network. Our paper introduces the first harmonized measures of the investment network across the development spectrum and documents novel empirical regularities, including systematic changes in the network with development. We propose a simple theory linking these disparities, which are endogenous to the path of sectoral prices, to differences in income per capita across countries. We show that the elasticity of output to sectoral productivity depends on the investment network, is highest in the Construction sector in developing countries, and in the ICT and Service sector in developed countries. Forces related to structural change, sectoral productivity and shifts in sectoral linkages drive this heterogeneity. For our sample of 58 countries, we show that 28% of cross-country differences in income per capita can be accounted for by disparities in the investment network. These differences are twice as large as the role of capital in income disparities obtained from standard development accounting.

Keywords: Investment Network, Structural Change, Growth, Trade in Capital.
JEL classifications: E23; E21; O41.

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†Columbia University; lc4053@columbia.edu
‡Cornell University and CEPR; julieta.caunedo@cornell.edu

1 Introduction

Capital accumulation and the systematic reallocation of economic activity across sectors are two of the most salient features of economic development. Different sectors utilize various investment goods for production, which are either produced by other sectors of the economy or imported. As economic activity shifts across sectors, the economy's ability to produce new capital—or to export goods in exchange for this new capital—changes, facilitating further capital accumulation. The study of the nature of this continuous feedback is crucial for understanding the mechanics of economic development (Hirschman, 1958). Such a study requires measures of sectoral links in both the production and use of new capital, i.e., the investment network. This paper provides the first harmonized measures of investment networks across the development spectrum. We document novel facts about the characteristics of these networks and construct a theory to evaluate the impact of these differences on observed income disparities across countries.

Recent studies have documented systematic changes in the sectoral composition of inputs used for investment as countries develop, Garcia-Santana, Pijoan-Mas and Villacorta (2021); Herrendorf, Rogerson and Valentinyi (2021). Work has also examined how disparities in the bundles of capital goods used for sectoral production lead to structural change (Caunedo and Keller, 2023). We combine these two approaches and characterize the link between production and uses of capital, i.e. the investment network, and aggregate output. Through the lens of a neoclassical multisector open economy model, we show that the investment network is a key component of the elasticity of GDP to sectoral productivity. The extensive literature in networks labels this elasticity as "*sectoral influence*", Acemoglu, Carvalho, Ozdaglar and Tahbaz-Salehi (2012).

Sectional influence is a function of the input-output structure and the investment network, summarizing the direct and indirect impact of changes in sectoral productivity on aggregate economic activity. Our economy has several distinct features relative to other economies with sectoral linkages. First, welfare and GDP differ due to the presence of investment. Along a growth path with constant investment rates these two are proportional to each other. Second, Domar weights depend on equilibrium investment rates and therefore on the entire equilibrium path in an economy that displays non-trivial transition dynamics. Since welfare is a Domar-weighted sum of sectoral productivities, welfare also depends on the entire equilibrium path. This feature is a consequence of the durable nature of capital and a novel result in the production networks literature that, for the most part, focuses on nondurable inputs. Third, *sectoral influence* is independent of the investment rate and depends solely on the characteristics

of the IO and investment network at each point in time. The reason is that the effect on aggregate GDP includes both consumption and investment, while the effect on welfare includes the trade-off between current and future consumption.

To bring empirical content to the roundabout effects summarized in *sectoral influence*, one needs measures of intermediate and investment networks, as well as factor shares and investment rates. While estimates of the input–output structure have become increasingly available across countries, estimates of the investment network are only available for the US (vom Lehn and Winberry, 2022), and a handful of years across OECD economies.[1] We advance previous measurement efforts by providing cross-country and time-series harmonized estimates of the investment network for 58 countries with income per capita levels between $428 and $81,599 constant 2015 PPP dollars and time spans that date back to the 1960s for a subset of countries. In our analysis, capital is disaggregated into multiple equipment types, including ICT, electronics, machinery, transportation and other durables, as well as structures, measured through construction investment.[2]

To create our harmonized measures, we exploit a methodology similar to that of the Bureau of Economic Analysis (BEA) in the US. The BEA combines the occupational composition of each industry and an allocation rule for capital to workers to estimate investment by capital type and sector (Meade, Rzeznik and Robinson-Smith, 2003). Unfortunately, the apportioning of stocks to workers is not publicly available. Hence, to ensure replicability, we opt for an allocation of capital across sectors that follows Caunedo, Jaume and Keller (2023a) for equipment sectors and an allocation that follows intermediate inputs for construction and other sectors with positive investment in the national accounts. Our own estimates of the investment network for the US closely follow those published by the BEA, which yields confidence on our apportioning method. Still, the BEA and our apportioning depends on sectoral occupational compositions to impute capital uses. Ideally, one would like to have direct measures of investment by detailed categories in each sector. This data is broadly unavailable even within a country, let alone across countries and for long time series. There is a handful of countries with VAT data, that allow for measurement of firm-to-firm transactions, notably Chile between 2015-2024 Gillmore, Kohn, Heresi, Quevedo and Rivera (2025). We find that our estimates are fully consistent with VAT estimates which again is reassuring given the widely different sources of these estimates.

[1] These measures are self-reported by country offices to the OECD Statistics office, and it is unclear whether measurement is comparable across countries. Ding (2023) exploits these investment flows to estimate capital services in each sector from different sectors and countries. To do so, he uses bilateral import flows to input cross-country linkages and estimates user costs along a BGP. In other words, he treats the investment network as a primitive.

[2] Our benchmark estimates include 10 sectors, but estimates for as many as 19 sectors consistently defined across countries and time can be made readily available.

With our newly constructed measure, we document some novel facts. First, there are systematic changes in the composition of investment providers as countries develop, which we measure through *outdegrees*—the row sums of the network. Construction is the dominant provider at all income levels and follows a hump-shaped relationship with development; other services (including financial intermediation and real estate) are also major suppliers, but their importance declines with income and is overtaken by a strong increase in ICT, whose role nearly triples as countries get richer. Second, the bundle of capital goods that sectors use in production becomes more diversified as countries grow richer, with the exception of construction, where concentration rises. Finally, the investment network and the input–output tables differ substantially: sectors that are important providers of intermediate inputs are not necessarily those that matter for investment supply. This is the case of agriculture—a major provider of intermediate inputs (especially in low-income countries) that plays only a minor role in investment—and of construction, which is more than three times as important a provider of investment as of intermediate inputs. ICT is an important provider of both investment and intermediate goods. Moreover, homophily (the diagonal weight) is stronger in the input–output network, meaning sectors are more important providers of intermediate inputs to themselves than of investment goods. Hence, sectoral productivity shocks have different aggregate implications depending on whether a sector mainly supplies intermediates or investment.

A natural question to ask is whether sectors with high *outdegrees* in the investment network are also sectors where changes in productivity have the strongest impact on aggregate activity. The answer is no. Our theory predicts that their role is mediated by other features of the economy, including value added and capital shares in production, as well as the full input–output network. Hence, to assess the role of the investment network for income disparities, we exploit the characterization of *sectoral influence*.

Sectorial influence is an endogenous outcome of the model because it depends on endogenous value added shares, as well as the sectorial expenditure shares in investment which are functions of relative prices. We conduct two sets of exercises to assess the role of the technology for producing investment for income disparities. One exercise in partial equilibrium and another one in general equilibrium.

First, we conduct standard income accounting exercise in partial equilibrium, i.e. where the value added shares that enter into sectorial influence are taken exogenously. Given measured roundabout effects as summarized in an “augmented" Leontief inverse (augmented by investment linkages), we can calibrate sectoral productivity differences to match disparities in value added across sectors within countries and across countries for a given sector. Our main

exercise is then to compare our baseline economy, which by construction exactly matches income disparities across countries, to an economy where the investment network consists of a series of stacked columns equal to the sectorial value added shares. This counterfactual economy correspond to an economy with a single capital good (rather than sector-specific ones) which is produced with the same technology as final consumption. We show that 15% of the variation in observed income disparities is accounted for an investment technology that differs from that of the one-capital economy. This 15% is an additional contribution of the technology for producing capital beyond what measured capital disparities would dictate (in our sample, measured capital disparities account for 13% of income differences). Hence, the role of capital is 28% of the observed cross-country disparities in income. We conclude that accounting for roundabout effects in the technology for producing capital almost doubles its role for income disparities. In this partial equilibrium set up we also study the role of imported equipment for these results. The imported nature of capital is potentially relevant to our accounting because the magnitude of the roundabout effect is sensitive to how much capital is locally produced and because countries may buy productive capital by generating exports. Our theory accommodates these channels by modeling investment goods as a combination of locally and foreign-sourced investment and by explicitly modeling exports of final goods as a source to finance investment imports. We run counterfactual exercises where we construct the loadings of the investment network including only the domestically produced share of investment and shut down the productivity amplification through the production of tradable goods. We find that the counterfactual economy induces a 44% decline in steady-state income per capita on average across countries and that the variance of income falls by 12% relative to baseline. In other words, trade accounts for slightly more than one-third of the variance in income accounted for by the investment network.

Second, we reassess the role of the technology for producing capital in an economy where both the size of the sectors (VA shares) and the investment network are determined in equilibrium through differential rates of sectorial productivity growth, differential intensity in the use of sectorial investment for production and income effects. We solve the transition path in a simplified economy with five sectors, including agriculture, manufacturing, services, construction and ICT. We calibrate our economy to the path of development of Korea between 1965 and 2014 to target the path of sectorial relative prices and consumption shares, as well as the average level of value added shares in the economy. Despite not targeting trends in value added shares as well as the features of the cost shares of investment across sectors, the model does a remarkable good job at tracking structural change in value added, as well as the shifts in the investment network as Korea developed. We then run counterfactual exercises that provide

the general equilibrium counterpart to the partial equilibrium exercises. We run counterfactuals where we change sectorial productivities one-at-the time and evaluate sectorial influence in general equilibrium. We find relatively low role for productivity growth in ICT, except possibly for value added in the service sector, and a very important role for productivity growth in the manufacturing sector, as a key driver of improvements in aggregate output levels (shrinking to a quarter of its baseline growth) and capital output levels, shrinking by a factor of 3.

Contribution to the literature. There is a growing literature studying the relevance of sectoral linkages, mostly through intermediate input use, for differences in income per capita across countries; see Ciccone (2002); Jones (2011). This role is quantified in Fadinger, Ghiglino and Teteryatnikova (2022), who show that differences in the input–output structure amplify the role of sectoral productivity for differences in income per capita. We document that the input–output and investment networks are empirically different, leading to differential roles in driving income disparities across countries. We show that disparities in the technology for producing investment and consumption goods amplify or dampen the effect of sectorial productivity differences for income disparities. In other words, standard development accounting exercises Hall and Jones (1999) are misspecified when the technology for producing investment and consumption differ. Quantitatively, we find that disparities in measured capital are more important to income disparities than previously understood.

Garcia-Santana *et al.*, 2021; Herrendorf *et al.*, 2021 document systematic shifts in the inputs used for investment as economies develop. In the US time series, Gaggl, Gorry and vom Lehn (2023) document disparate shifts in the composition of inputs for investment and consumption, while Caunedo and Keller (2023) show how differences in the investment bundles used by different sectors drive sectoral reallocation. Our paper completes the puzzle by documenting disparities in the investment bundle and the composition of investment along the development spectrum. The study of the role of the investment network in determining GDP is relatively recent. In the US, vom Lehn and Winberry (2022) focus on short-run fluctuations, while Foerster, Hornstein, Sarte and Watson (2022) study implications for long term growth. [3]

Another novel finding of our theory is that sectoral investment rates mediate the impact of sectoral productivity on aggregate GDP and affect the rate of convergence of the economy to its (asymptotic) BGP. As in Liu (2019), output elasticities to sectoral productivity are different from Domar weights. In our model, this difference arises from the presence of investment, i.e., sectoral value-added shares and consumption shares do not equalize. Indeed, because our economy is efficient, welfare is a Domar-aggregated measure of sectoral productivities.

[3] Differently from Foerster *et al.* (2022), we solve for the endogenous allocation of labor, our network changes endogenously with shifts in sectoral relative prices and we allow for equipment imports.

These Domar weights are a function of equilibrium investment rates, a channel that is absent in economies where inputs are nondurable, as in Acemoglu *et al.* (2012) and the extensive literature that follows. Buera and Trachter (2024) is the closest paper to our study, characterizing differential GDP responses to sectoral productivity. Their work emphasizes the role of distortions and nonconvexities for endogenous technology choices. Our paper instead emphasizes the intensive margin of the adoption of capital-embodied technology, i.e., investment, and documents systematic disparities in its nature along the development spectrum.

The remainder of the paper is organized as follows: Section 2 discusses an illustrative example for the role of the investment network in accounting for income disparities. Section 3 discusses the methodology to construct estimates of the investment networks and novel empirical regularities of the investment network; Section 5 presents the model and Section **??** discusses the main result from income accounting exercises while Section 6.1 documents patterns of *sectoral influence* along the development spectrum. Section 8 concludes the paper.

2 An illustrative example

We begin with a simple economy in which the technology for producing new capital differs from that for producing consumption goods in its input requirements. Consider a two-sector economy with the following technologies for sectoral output, capital accumulation, and market clearing:

$$v_{it} = \exp(z_{it}) K_{it}^{\alpha} L_{it}^{1-\alpha}$$

$$K_{it+1} = (1-\delta) K_{it} + \prod_j x_{jit}^{\omega_j}$$

$$c_{it} + x_{it} = v_{it} \qquad x_{it} = \sum_j x_{ijt}$$

Sectoral output may be consumed or invested. Sectoral output itself is produced using sector-specific capital and labor via a Cobb-Douglas technology with heterogeneous productivity across sectors. Investment goods are assembled from sectoral inputs, with each sector j entering the investment aggregator with intensity ω_j—a weight that is common across all sectors that use input j. We relax both this symmetry and the Cobb-Douglas assumption later in the paper.

Because the investment aggregator is identical across sectors, the price of capital is equalized, which in turn equalizes capital-labor ratios: $k_i/l_i = K/L$. Log GDP is then an expenditure-share-weighted sum of log sectoral value added, $\ln(v_t) = \sum_{i=1,2} \tilde{\eta}_{it} \ln(v_{it})$.

The equalization of capital-labor ratios, combined with inelastic labor supply, implies that GDP takes the simple form $\ln(v_t) = \sum_{i=1,2} \tilde{\eta}_{it}(z_{it} + \ln(\tilde{\eta}_i)) + \alpha \ln(K_t)$. If agents have no disutility from labor and derive utility from sectoral consumption in a log-separable manner, the economy admits a steady state in which output can be written as a function of the capital-output ratio:

$$\ln(v) = \frac{1}{1-\alpha} \sum_{i=1,2} \tilde{\eta}_i(z_i + \ln(\tilde{\eta}_i)) + \frac{\alpha}{1-\alpha} \ln(K/v).^{4}$$

This expression mirrors that of the standard one-sector growth model, with aggregate TFP equal to a weighted average of sectoral productivities. For a given capital-output ratio, the pass-through from sectoral productivity to output is governed by expenditure shares, amplified through capital accumulation by the factor $1/(1-\alpha)$.

In the canonical one-sector model, the capital-output ratio is independent of TFP in steady state, which allows one to hold the marginal product of capital fixed when varying productivity—cleanly separating the contributions of technology and capital to cross-country income differences (Klenow and Rodríguez-Clare, 1997; Hall and Jones, 1999). In our economy, this separation breaks down. The marginal product of capital depends directly on sectoral productivities:

$$\text{MPK} \equiv \alpha\omega_i \frac{v}{K} \exp(z_n)^{\omega_n - \tilde{\eta}_n} \exp(z_i)^{\omega_i - \tilde{\eta}_i} \frac{\omega_n^{\omega_n} \omega_i^{\omega_i}}{\tilde{\eta}_n^{\tilde{\eta}_n} \tilde{\eta}_i^{\tilde{\eta}_i}}.$$

The marginal product of capital is independent of sectoral productivity—and therefore the equilibrium capital-output ratio is pinned down by technology alone—only when the GDP and investment aggregators coincide. Section 3 documents empirically how these two technologies differ, allowing additionally for variation in investment intensities across sectors, $\omega_{ij} \neq \omega_{jj}$, and over time.

The dependence of the marginal product of capital on productivity has an important consequence: standard development accounting cannot isolate the direct effect of technology from its effect on capital accumulation. Crucially, the direction in which productivity improvements affect aggregate capital is ambiguous, depending on how intensively a sector's output is used in producing investment versus consumption. When productivity rises in a sector that contributes disproportionately to investment production—$\omega_j > \tilde{\eta}_j$—the capital-output ratio increases. When it rises in a sector whose output is used predominantly for consumption, such as services—$\omega_j < \tilde{\eta}_j$—the capital-output ratio falls.

It follows that applying a misspecified one-sector framework, when investment and consumption technologies genuinely differ, will either overstate or understate the role of technol-

[4] The term $\ln(\tilde{\eta}_i)$ captures the static allocation of labor.

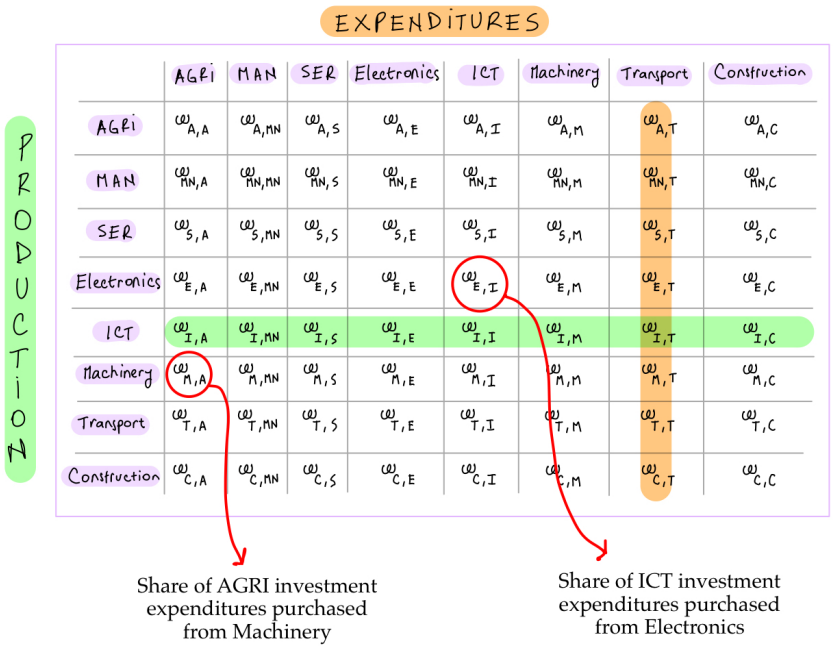
ogy depending on the investment intensity of the sectors in which productivity grows. If productivity gains are concentrated in sectors that supply investment goods disproportionately, one-sector accounting will underestimate the contribution of technology and overestimate that of capital. If gains are concentrated in consumption-intensive sectors, the bias runs in the opposite direction. As we argue in Section 6, the cross-country evidence points to the latter case.

3 Building cross-country measures of investment networks

In this section we describe how we construct estimates of the investment technology along the income spectrum. This amounts to estimating expenditure shares in new capital, by capital type and purchasing sector, across countries and over time.

3.1 Methodology

Figure 1: Example: Allocation of ICT Investment Flows



Notes: Illustrative example of the investment network. Columns indicate consuming sectors, while rows indicate production sectors. Entries are expenditure shares by consuming sectors in different investment types, ω.

Figure 1 presents an example of an investment network. Each entry (i, j) records total investment expenditures by column sector j on goods purchased from row sector i. Summing across columns yields total investment production by each row sector; summing across rows yields total investment expenditures by each column sector. For instance, each element of the ICT row shows how much investment the ICT sector supplies to each purchasing sector j, while each element of the agriculture column shows how much investment agriculture purchases from

each supplying sector i. Expenditure shares ω_{ij} are obtained by dividing each entry in column j by that column's total investment expenditures, so that column shares sum to one: $\sum_i \omega_{ij} = 1$.

Estimates of investment produced by each sector—that is, the row sums in Figure 1—are readily available from *use tables*, which record how sectoral output is allocated across intermediate and final uses, including consumption and investment. Importantly, our measure of investment flows encompasses both domestically produced and imported equipment; see Section 3.2. For ease of exposition, we refer to these flows as "produced" by a sector, though the category includes imports.

Our contribution is to develop an imputation strategy for determining how much of the investment produced (or imported and allocated to investment) by each sector is purchased by other sectors of the economy. Our approach follows the BEA's procedure for the US wherever it is replicable and wherever the required information is available across countries and over time. Meade *et al.* (2003) describes three stages of this imputation:

1. direct assignment;
2. allocation proportional to sectoral occupational composition;
3. allocation proportional to sectoral capital expenditures (used only for structures).

Direct assignment places all investment in a given category into its most likely use—for example, nuclear plant investment is assigned to the utilities sector. The second method assumes that a sector's occupational mix is informative about the types of capital it employs. For structures, the BEA allocates investment in proportion to sectoral capital expenditures.

To maximize replicability, we forgo manual assignment of investment categories to user sectors (method 1), as this requires finely disaggregated data that are not consistently available internationally. We adopt method (2) as our benchmark and apply a variant of method (3) when (2) is infeasible; in particular, we allocate expenditures in proportion to each sector's demand for intermediate inputs.[5]

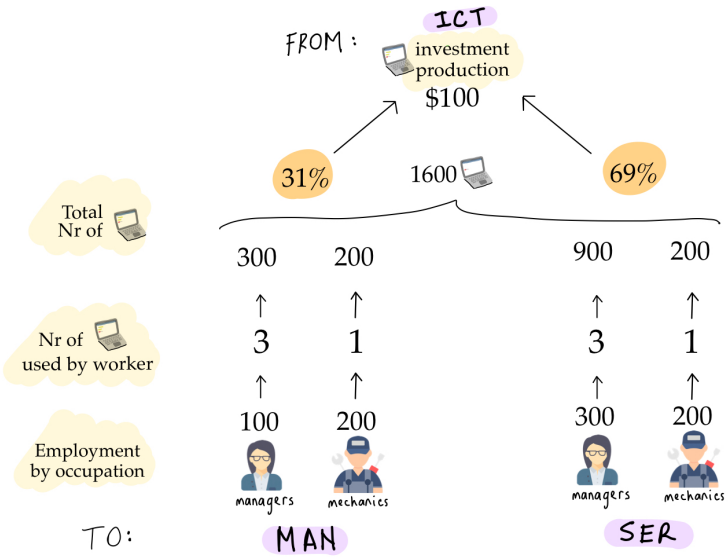
3.1.1 Allocation of equipment investment flows

We allocate equipment investment flows primarily according to the occupational composition of each purchasing sector and the types of capital those occupations are likely to use.[6]

[5] The BEA typically uses micro-data on structures investment to allocate those flows, but such data are not available for our cross-country, cross-sector, and time-series sample.

[6] The BEA's allocation follows its public documentation; however, the details of how capital is assigned to specific occupations and sectors are not publicly available. In Section 4.5, we validate our methodology by comparing our U.S. estimates with those of the BEA.

Figure 2: Example: Allocation of ICT Investment Flows



Notes: Illustrative example of the allocation of investment flows across purchasing sectors, obtained by exploiting the sectoral occupational composition and the equipment intensity across occupations from Caunedo *et al.* (2023a).

Figure 2 illustrates this assignment for ICT investment. Suppose there are three sectors—ICT, manufacturing, and services—and the ICT sector produces $100 of new capital goods (software). Our goal is to determine how much of this $100 is purchased by manufacturing and services respectively. To do so, we use the occupational mix of each purchasing sector together with the relative intensity with which those occupations use software.

In the example, manufacturing and services each employ 200 mechanics; manufacturing also employs 100 managers and services 300 managers. Managers use software three times as intensively as mechanics, so for every dollar assigned to mechanics, three are assigned to managers. These relative-use intensities come from the tool–occupation allocation in Caunedo *et al.* (2023a) (details below). It follows that for the manufacturing sector as a whole, out of every $500 of software, $300 is used by managers and $200 by mechanics. For the services sector, the corresponding allocation out of $1,100 is $900 to managers and $200 to mechanics. Rescaling these relative magnitudes to the $100 of software produced by ICT, we assign 31% to manufacturing and 69% to services.

The software example assumed a one-to-one mapping between equipment types and producing sectors. While this holds for some equipment types, a given type of equipment is often produced by several sectors. This information is encoded in "bridge tables" that underlie the national accounts. For example, using the average U.S. bridge table from 2000 to 2018, 78% of computer investment is produced by the electronics sector and 22% by the ICT sector; conversely, 28% of electronics output consists of computers, 34% of communication equipment,

and the remainder of other equipment categories.

Our methodology therefore proceeds in three mapping steps: (i) from investment-producing sectors to equipment types, using bridge tables; (ii) from investment-purchasing sectors to equipment types, using the occupational distribution and the tools used on the job; and (iii) combining (i) and (ii) to map investment-producing sectors to investment-purchasing sectors—that is, to construct the investment network.

From investment-producing sectors to equipment types. Bridge tables are not, to the best of our knowledge, available for countries other than the United States. We therefore use the average allocation of sectoral production to equipment types from U.S. bridge tables covering 2000–2018.[7] The dollar value of investment in equipment type e produced by sector i in country c at year t is obtained by multiplying the U.S. bridge-table share b_{ie}^{US} by the total investment produced (or imported and allocated to investment) by sector i in country c at year t, x_{it}^{c} (from use tables):

$$x_{iet}^{c} = x_{it}^{c}\, b_{ie}^{\text{US}}. \tag{1}$$

From investment-purchasing sectors to equipment types. We assign equipment investment to purchasing sectors using the tools employed within each occupation in the United States, as described by O*NET. Specifically, we implement the methodology of Caunedo *et al.* (2023a), which assigns equipment investment—and therefore equipment stocks—to workers in different occupations.[8] As illustrated in Figure 2, the key identifying assumption is that the relative intensity with which a manager uses a given piece of equipment compared to a mechanic is the same across countries and equal to the U.S. value.[9] The absolute investment allocated to each purchasing sector still varies across countries because ICT investment production differs internationally and because the occupational composition of industries varies with the level of development.

We first compute the share of total production of equipment type e purchased by industry j in country c at time t:

$$\gamma_{ejt}^{c} = \frac{\sum_{o} \tau_{e}^{o,\text{US}}\, n_{jot}^{c}}{\sum_{j}\sum_{o} \tau_{e}^{o,\text{US}}\, n_{jot}^{c}}, \tag{2}$$

[7] A sector's total production allocated to equipment types excludes replacement of used goods and trade margins. When we impute investment flows in other countries—which may include these margins—we effectively distribute them equally across equipment types.

[8] The methodology in Caunedo *et al.* (2023a) crosswalks equipment categories to the tools used within each SOC occupation. We use Dingel and Neiman (2020)'s crosswalk between SOC and ISCO to map these tools to harmonized cross-country occupational definitions.

[9] As a robustness check, we construct country-specific tool intensities within occupations by combining cross-country within-occupation variation in tasks with a projection of tool usage onto task intensity. This variation accounts for roughly one third of the cross-country variation in the investment network; see Appendix B.6.

where n^c_{jot} is the number of workers in occupation o and industry j in country c at time t, and $\tau^{o,\text{US}}_e$ is the intensity with which a worker in occupation o uses capital type e in the United States. Since γ^c_{ejt} distributes a given equipment type across industries, the shares sum to one: $\sum_j \gamma^c_{ejt} = 1$.

We then estimate the dollar value of investment in equipment type e purchased by sector j in country c at year t as the product of the value of investment in type e produced by sector i (equation 1) and the purchasing share in equation 2:

$$x^c_{ejt} = x^c_{iet}\, \gamma^c_{ejt}. \tag{3}$$

From investment-producing sectors to investment-purchasing sectors. For each purchasing sector j, we aggregate across all equipment types produced by sector i to obtain the total dollar value of investment that j purchases from i:

$$x^c_{ijt} = \sum_{e \in \mathcal{E}_i} x^c_{ejt}, \quad \text{for } i \in \text{equipment sectors}, \tag{4}$$

where $\mathcal{E}_i$ denotes the set of equipment types produced by sector i.

The investment network. Investment expenditure shares ω^c_{ijt}, which constitute the investment network, are defined as:

$$\omega^c_{ijt} = \frac{x^c_{ijt}}{\sum_i x^c_{ijt}}. \tag{5}$$

3.1.2 Allocation of non-equipment investment flows

For investment flows originating in the construction, agriculture, nondurables, transportation services, and other services sectors, no information on worker usage of capital is available. We therefore allocate these flows in proportion to each sector's role as a supplier of intermediate inputs to other sectors, drawing on information from input-output tables.

Let $\tilde{\mu}^c_{ijt}$ denote the share of total intermediate inputs produced by sector i that are purchased by sector j. For non-equipment sectors, the dollar value of each entry in the investment table is:

$$x^c_{ijt} = \tilde{\mu}^c_{ijt}\, x^c_{it}, \quad \text{for } i \notin \mathcal{E}_i.$$

Investment expenditure shares are then computed as in equation 5.

3.2 Data Description

Based on the methodology described in the previous section, constructing the investment network requires five pieces of information: (i) sectoral production of investment goods, x_{it}^c; (ii) tools per worker by occupation, $\tau_e^{o,\text{US}}$; (iii) employment by occupation and sector, n_{iot}^c; (iv) a bridge table mapping sector i to equipment type e, b_{ie}^{US}; and (v) the input–output structure, $\tilde{\mu}_{ijt}^c$. Table 4 summarizes the sources, and Appendix Table 5 provides a detailed description.

Investment production by sector and input–output tables. We obtain gross fixed capital formation (GFCF) by sector from the *use tables* that underlie the input–output accounts. The *use tables* report investment production by sector, whether produced domestically or imported. For our baseline investment network in each country, we use total investment production by sector (domestic plus imported).[10] We also construct a domestic investment network for each country based on domestic investment production by sector. For the nine Sub-Saharan African countries in our sample, we use data from Mensah and de Vries (2023). For the remaining countries, we use the World Input–Output Database (WIOD) and the OECD input–output tables. We map equipment production into sectoral investment production using the U.S. bridge tables (as explained before).

Employment by occupation and sector. We use employment by occupation and sector from PIAAC, IPUMS International, and ILOSTAT. For countries with data from all three sources, we favor PIAAC over IPUMS International and ILOSTAT because PIAAC uses more detailed occupational categories.[11] In IPUMS International, the harmonized industry classification does not disaggregate equipment sectors within manufacturing; we instead used the raw datasets which include the entirety of the industry classification available in the labor survey. For IPUMS countries we manually construct crosswalks between the disaggregated (non-harmonized) industries and our 10 sectors. In PIAAC and ILOSTAT, sectors are classified according to ISIC Rev. 4 or ISIC Rev. 3, which we also crosswalk to our 10-sector scheme (see online appendix).

We use 2-digit occupational categories to assign relative intensity of tools by equipment type and then aggregate them up to 1-digit to have a common occupational classification across countries. We assign tools to workers in different occupations following the intensity measure in Caunedo *et al.* (2023a).

Country coverage. Our dataset covers 58 countries at different stages of development, with

[10] GFCF flows are reported in nominal currency; we deflate them using output PPP prices from the Penn World Table. To abstract from business-cycle fluctuations, we apply a Hodrick–Prescott (HP) filter to sectoral GFCF flows.

[11] PIAAC measures aggregated at the 1-digit level correlate strongly with IPUMS data; see Caunedo, Keller and Shin (2023b).

GDP per capita (PPP) ranging from $428 to $81599. For 20 of these countries, we construct time series of investment networks from 1965 to 2014; for the 9 countries in Sub-Saharan Africa, from 1990 to 2019. For the remaining 29 countries, the investment-network time series covers 2000–2014. See Table 5 in the Appendix for a full description.

4 The Investment Network along the Development Spectrum

We start by showing how the composition of sectors that provide investment changes with development. Then, we show how the investment network diversifies with income, and how it differs from the input-output network.

4.1 Investment Network Outdegrees

We begin by documenting each sector's *outdegree*—the row sum of the investment network. In Table 1, we group countries by the World Bank income classification and report the average sectoral outdegree for each group.[12] Several patterns emerge. Construction is the dominant supplier of investment goods across the distribution and follows a hump-shaped relationship with development. Other services—including financial intermediation and real estate—are also major suppliers, but their importance declines with income. The sector with the strongest increase is ICT: high-income economies exhibit outdegrees nearly three times those of low-income economies. Transportation equipment also rises with income, though more moderately. Electronics and transport services display hump-shaped profiles, whereas durables and nondurables fall from low to middle income and then tick up slightly at high income. Finally, Agriculture is a minor supplier in all groups, and its importance diminishes sharply with development.

Table 1: Investment Network Outdegrees

Notes: Data for 2005, for low-, middle-, and high-income countries (World Bank classification). Outdegrees are sectoral row sums in the investment network. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

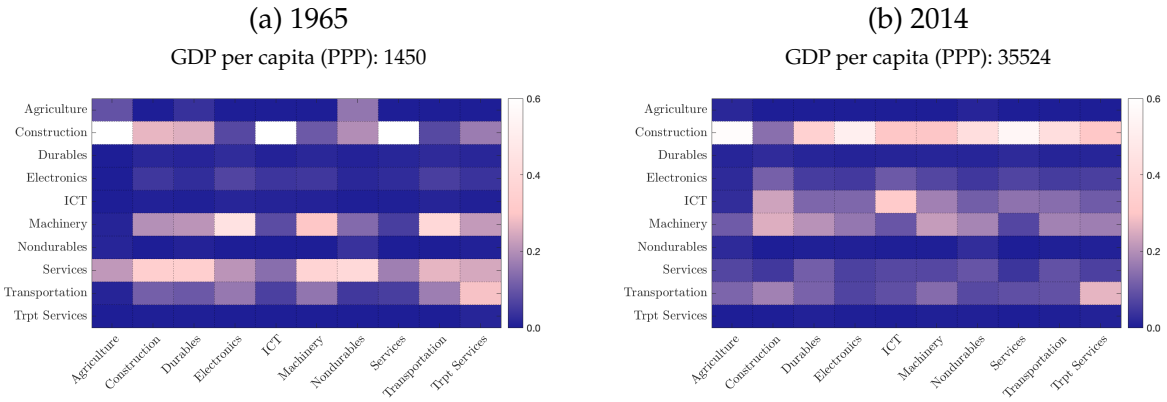
Similar patterns emerge when we examine South Korea's evolution from the low-income group to the high-income group—i.e. from 1965, when GDP per capita (PPP) was 1450, to 2014, when income per capita increased 24-fold. Figure 3 depicts the investment network in both years. Comparing panels (a) and (b), the same patterns as in Table 1 hold: construction is a prominent supplier of investment goods at all stages of development; agriculture is a minor

[12] Country lists appear in Table 7. For cross-country comparisons we use 2005, the year with PPP deflators; results are similar with other reference years or when pooling all country–year observations.

supplier; the role of ICT rises with income; and the importance of other services declines with development.

The investment network of Korea also becomes more diversified as the country grows richer, in the sense that the bundle of investment goods that a sector uses becomes more even across producing industries. This diversification is not specific to South Korea, it also characterizes the cross-section of countries along the development spectrum.

Figure 3: South Korea: Investment Network Over Time



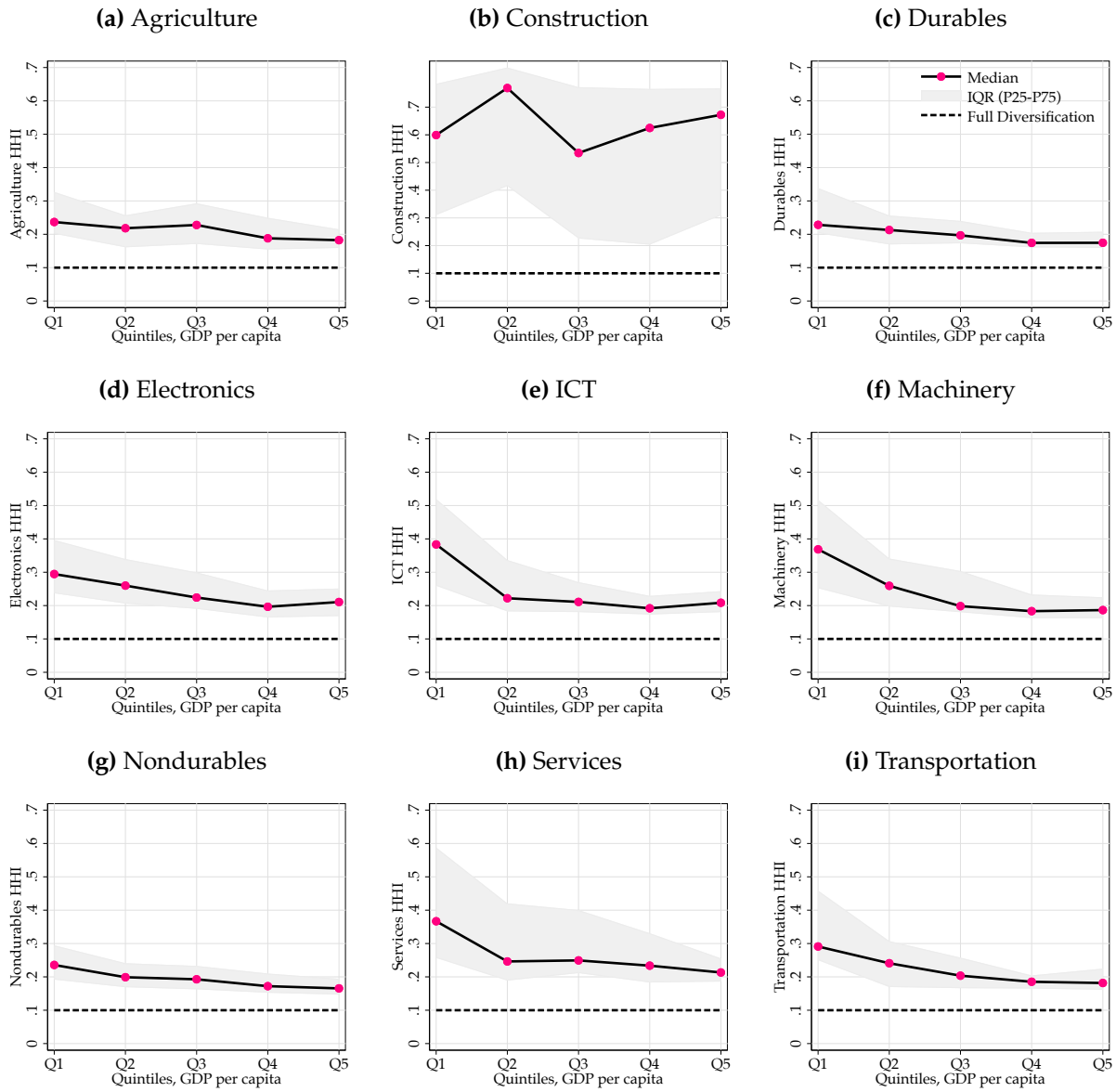
Notes: Estimates of South Korea's investment network over time. Each cell reports the investment-expenditure share of the *column* sector, as defined in Figure 1.

4.2 Investment Network Diversification

Figure 4 plots the Herfindahl–Hirschman Index (HHI) for each (column) sector that uses capital across quintiles of income per capita. Lower values indicate that a sector has a more diversified bundle. How does this structure change with development? For most sectors, the bundle of capital goods becomes more diversified as countries grow richer, moving closer to a fully diversified benchmark. The exception is construction, where concentration rises (about 56% from low to high income), largely because construction increasingly sources capital goods from its own sector. Diversification gains are most pronounced in capital-intensive activities—electronics, ICT, machinery, services, and transportation. In ICT, for instance, the HHI falls by roughly one-half (from about 0.45 in low-income to 0.19 in high-income countries), approaching the 0.10 value associated with full diversification.[13]

[13]Table **??** reports, for each sector, the mean and the 25th and 75th percentiles of the HHI for the low-, middle-, and high-income country groups.

Figure 4: Investment Network HHI



Notes: Data for 2005. The concentration index is the Herfindahl–Hirschman Index (HHI), computed as the sum of squared shares in each *column* of the investment network. Panels report medians and interquartile ranges of sectoral HHIs across quintiles of GDP per capita. The dashed line denotes the equal-split (full-diversification) benchmark, where investment is sourced equally from all capital types.

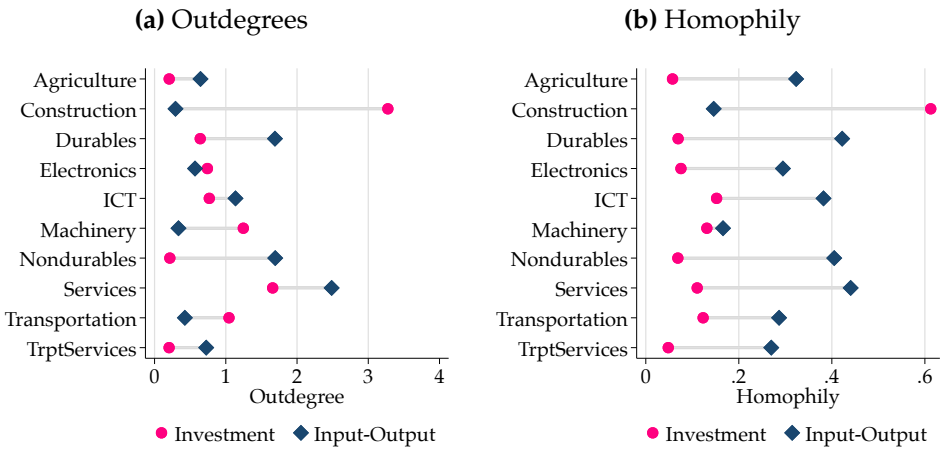
4.3 Investment Network vs. Input–Output Network

While the characteristics of the investment network were not known prior to this paper, there is considerably more evidence about the features of the input–output structure (see, for example, Fadinger *et al.* (2022)). Figure 5 documents differences between these two networks for the average country, as measured by (a) outdegrees, and (b) homophily. Moreover, we report these statistics in the Appendix for countries at different stages of development.[14]

Outdegrees. Several patterns stand out. Agriculture is a major supplier of intermediate

[14]Figure 15, reports the same statistics for low, middle, and high income country groups.

Figure 5: Investment Network vs. Input–Output Network



Notes: Data for 2005, averaged across countries. Circles denote the investment network; diamonds denote the input–output network. Panel (a) reports sectoral outdegrees (row sums); panel (b) reports sectoral homophily (diagonal elements of each network).

inputs—especially in low-income countries—but a minor provider of investment. Construction has the highest investment outdegree yet is among the least important intermediate input suppliers; its investment outdegree falls slightly with income, while its input–output outdegree rises. ICT outdegrees increase with development in both networks, with a large gap in low-income countries that narrows at high income. Electronics and transport services are hump-shaped, and transportation is predominantly a supplier of investment goods. Other services (including transport services) and nondurable manufacturing mainly supply intermediates rather than investment.s

Homophily. Comparing diagonal entries across the two matrices reveals a clear pattern: sectors heavily self-supply *intermediate* inputs (high input–output diagonals) but rely more on other sectors to source *investment* goods (lower investment-network diagonals). Two exceptions stand out: construction, which contributes little to its own intermediates but is a major supplier of investment goods to itself, and machinery, which shows similar homophily in both networks. These patterns hold across the income distribution and within countries over time.[15]

4.4 Sources of Cross-Country Variation in Investment Networks

The investment network for each country is constructed from two ingredients: sectoral investment flows and the occupational composition of the labor force. A natural question is how much of the cross-country variation in investment networks documented above is driven by differences in investment flows versus differences in employment composition. To answer this,

[15]Figure 16 compares Korea in 2014 and shows that diagonal loadings are systematically lower in the investment network than in the input–output network.

we construct two counterfactual investment networks for each country.

The first counterfactual replaces each country's occupational employment shares with those of the United States, keeping sectoral investment (GFCF) flows at their country-own levels. By equalizing the employment channel across countries, any remaining variation in the investment network is driven entirely by differences in investment flows. The second counterfactual does the reverse: it imposes U.S. GFCF levels on all countries while retaining each country's own occupational composition, so that residual variation reflects the employment channel alone. We assess the contribution of each channel both within countries—by computing the average distance between the baseline and counterfactual networks—and across countries, by comparing the cross-country variance of each element ω^c_{ijt} under each counterfactual to its baseline variance.

Differences in investment flows account for the bulk of cross-country heterogeneity in investment networks. When GFCF is equalized to U.S. levels, only 9% of the baseline cross-country variance in the typical element of $\mathbf{\Omega}$ remains—meaning that differences in the occupation composition explains little of the observed variation and investment flow differences explain roughly 91%. When occupational shares are equalized to U.S. levels instead, cross-country variance rises to about 130% of baseline, confirming that employment heterogeneity does not drive network dispersion—if anything, it slightly dampens it. Within countries, the GFCF channel plays a more important role for less developed economies relative to richer ones, consistent with their investment structures being furthest from the U.S. benchmark; the employment channel is broadly similar across all stages of development. Appendix Figure 17 and Table **??** report the full results.

4.5 Validation with other sources

Comparison with US Investment Networks. We compare our U.S. investment-network estimates to those constructed by vom Lehn and Winberry (2022) ("*VLW*"), which are based on the BEA capital-flows tables. We use 2012 as the primary reference year, as this is when data on employment by sector and occupation were collected by the PIAAC survey.[16] To control for potential disparities between BEA and WIOD in sectoral GFCF flows, we apply our methodology using the same sectoral investment-production estimates as in vom Lehn and Winberry (2022) (estimates are also robust to using WIOD investment flows). We then aggregate their 41-sector investment network to our 10 sectors, consistent with the ISIC Rev. 4 crosswalk in

[16]The Online Appendix shows that the patterns observed in 2012 are largely consistent with those from 1972 and 1992. Since occupational composition has changed over time, it is surprising that the assignment works relatively well 20 and 40 years earlier.

Table 6.

Appendix Table 14 compares estimates of (a) the sectoral outdegrees of the investment network—a measure of each sector's relevance as an investment provider to other sectors—and (b) the homophily of the investment network—a measure of each sector's relevance as a provider of investment to itself. The *outdegree* is the row sum of the investment network, and *homophily* summarizes the diagonal elements of the matrix. Comparing the estimates in the second and third columns of Table 14, panels (a) and (b), we find that our methodology aligns closely with vom Lehn and Winberry (2022), especially considering that the only common inputs are sectoral investment flows.

For further comparison, we regress the elements of our investment-network estimates on those from VLW and report the mean squared error (MSE) as a measure of fit. The MSEs are 0.005 for 2012, 0.008 for 1992, and 0.017 for 1972, indicating relatively small differences between our estimates and theirs.

Comparison with Chile's Investment Networks. A potential concern with ours and the BEA estimates of the investment network is that the allocation of investment to its uses depends on the occupational composition of the sector demanding equipment, rather than on information on actual investment in each sector. This data is rarely available. Chile provides a rare opportunity because of the availability of data on the universe of firm-to-firm transactions through the VAT system. We use investment networks estimates from Gillmore *et al.* (2025) based (on domestic) firm-to-firm transactions for comparison.

We compare our estimates with theirs through summary statistics of the network, including outdeegres and homophily, see Table 15. We find that for all sectors, except for ICT and Services our estimate of domestic investment networks align very well with their estimates from firm-to-firm transactions. The difference for services is that their methodology overestimates the effects of wholesale trade which are intermediaries of imported good and therefore show up as "traded" component in our series but domestic in theirs. The differences for ICT arises from disparities in the capitalization of certain investments, in particular software.

Having shown the key characteristics of the investment network along the development spectrum, we are now ready to analyze its importance for the development process. To do so, we build a model that rationalizes the distinctive features of the network, and its change with development, through differences in investment elasticities, sectoral productivities, and investment-specific technical change.

5 A Model of the Investment Network and Economic Development

The model is an open-economy version of Long and Plosser (1983), augmented to include a CES sectoral investment aggregator. Differences in aggregate GDP per capita across countries are associated with disparities in investment and input–output networks. The investment network adjusts endogenously to sectoral productivity trends.

5.1 Environment

The economy consists of N sectors that combine capital, labor, and intermediate inputs to produce output. Gross output in sector $n \in N$ is given by

$$y_{nt} = \left(\frac{\nu_{nt}}{\gamma_n}\right)^{\gamma_n} \left(\frac{m_{nt}}{1-\gamma_n}\right)^{1-\gamma_n},$$

where $\gamma_n \in [0,1]$ denotes the share of value added in gross output. Sectoral value added is

$$\nu_{nt} = \exp(z_{nt}) \left(\frac{k_{nt}}{\alpha_n}\right)^{\alpha_n} \left(\frac{l_{nt}}{1-\alpha_n}\right)^{1-\alpha_n},$$

and depends on sectoral TFP z_{nt}, capital and labor inputs k_{nt} and l_{nt}, and the capital share in value added α_n. Intermediate inputs are aggregated using a constant-returns-to-scale technology, $m_{nt} = \prod_{i=1}^{N} \left(\frac{m_{int}}{\mu_{in}}\right)^{\mu_{in}}$, where m_{int} denotes inputs from sector i used in sector n. The input shares μ_{in} define the input–output matrix M. Each column of the matrix M describes the composition of intermediate inputs used by a sector, with $\sum_i \mu_{in} = 1$, while each row captures the importance of a sector as a supplier of intermediates to the rest of the economy.

The capital stock used in each sector follows the law of motion

$$k_{n,t+1} = x_{nt} + (1-\delta_n)k_{nt},$$

where x_{nt} is a composite investment good for sector n. Investment is produced using inputs sourced from multiple sectors. Let χ_{int} denote investment goods from sector i used in the production of capital for sector n at time t. The investment aggregator is given by

$$x_{nt} = \exp(z^x_{nt}) \left(\sum_{i=1}^{N} \tilde{\omega}_{in}^{\frac{1}{\iota_n}} \chi_{int}^{\frac{\iota_n - 1}{\iota_n}}\right)^{\frac{\iota_n}{\iota_n - 1}}, \tag{6}$$

where $\tilde{\omega}_{in}$ are technological weights governing the importance of sector i in the production of investment for sector n, ι_n is the elasticity of substitution across investment goods, and z^x_{nt}

captures investment-specific technical change for sector n.

The allocation of investment expenditures across sectors is summarized by the investment network Ω_t, whose typical element is $\omega_{int} \equiv \frac{p_{int}\chi_{int}}{p_n^x x_{nt}}$. When the elasticity of substitution across investment goods is unitary, $\iota_n = 1$, these shares coincide with the technological weights, $\omega_{int} = \tilde{\omega}_{in}$. More generally, investment shares vary with relative prices. Rows of Ω_t capture the production of investment goods by each sector, while columns describe the composition of investment used by each sector. Since the investment technology exhibits constant returns to scale, column shares sum to one, $\sum_{i=1}^{N} \frac{p_{int}\chi_{int}}{p_n^x x_{nt}} = 1$.

Moreover, investment goods from sector i used by other sectors can be sourced domestically or imported: $\chi_{int} = \left(\frac{\chi_{int}^d}{1-\phi_i}\right)^{1-\phi_i} \left(\frac{\chi_{int}^f}{\phi_i}\right)^{\phi_i}$, where χ_{int}^d and χ_{int}^f denote, respectively, domestic and imported investment goods from sector i used by sector n, and ϕ_i is the import share in investment goods from sector i.

Output in each sector can be allocated to final goods c_{nt}, intermediate inputs, or domestic investment:

$$y_{nt} = c_{nt} + \sum_i m_{nit} + \sum_i \chi_{nit}^d.$$

Aggregate output Y_t is defined implicitly by the non-homothetic CES aggregator of Comin, Lashkari and Mestieri (2021),

$$\sum_{n=1}^{N} \theta_n^{\frac{1}{\sigma}} \left(c_{nt}\right)^{\frac{\sigma-1}{\sigma}} Y_t^{\frac{(1-\sigma)\epsilon_n}{\sigma}} = Y_t^{\frac{\sigma-1}{\sigma}}, \qquad \theta_n > 0,$$

where θ_n governs the loading of final goods from sector n, σ is the elasticity of substitution across goods, and ϵ_n is the sector-specific income elasticity.[17] Final goods can be allocated to exports, ϵ_t, or to consumption, C_t, by the representative household.[18] The household derives utility $U(C_t)$, which satisfies standard regularity conditions, and discounts the future at rate β. Final goods market clearing therefore implies

$$Y_t = C_t + \epsilon_t.$$

We consider a small open economy that exports final goods in exchange for capital goods of different types, as in Jones (2011). Net exports are defined as the difference between the value

[17] The induced expenditure shares take the form $s_{nt} \equiv \frac{p_{nt}c_{nt}}{p_{Yt}Y_t} = \frac{\theta_n p_{nt}^{1-\sigma} Y_t^{\epsilon_n - 1}}{\sum_{j=1}^{N} \theta_j p_{jt}^{1-\sigma} Y_t^{\epsilon_j - 1}}$. When $\epsilon_n = 1$ for all n, the aggregator collapses to standard CES and shares depend only on relative prices.

[18] Our findings are robust to allowing for separate aggregators for exports and consumption. When the price of consumption is defined in units of exports, the results are unchanged. Alternatively, exports can be specified as a constant fraction of final output Y_t.

of exports and the value of imports:

$$NX_t = p_{Yt}\epsilon_t - p_{\epsilon^f t}\epsilon_t^f.$$

The value of imports is given by the price index of imported goods multiplied by a composite import value, $\epsilon_t^f = \prod_{i=1}^{N}\left(\frac{\chi_{it}^f}{\phi_i^f}\right)^{\phi_i^f}$, as in Basu, Fernald, Fisher and Kimball (2005). The terms of trade are defined as the ratio of the price of exports to the price of imports, $\tau_t \equiv \frac{p_{Yt}}{p_{\epsilon^f t}}$, where the price of imported goods is a constant-returns-to-scale aggregator of the (exogenous) prices of imported investment goods.

5.2 Competitive Equilibrium

Definition 1 (Competitive Equilibrium)**.** *Given initial capital stocks* $\{k_{j0}\}_{j=1}^{N}$*, sequences of exogenous shocks* $\{z_{jt}, z_{jt}^x\}_{j,t}$*, and model parameters, a* competitive equilibrium *is a collection of sequences of allocations* $\{C_t, c_{jt}, x_{jt}, \chi_{ijt}, m_{jt}, m_{ijt}, l_{jt}, y_{jt}, k_{j,t+1}\}_{t\geq 0,\, i,j}$ *and prices* $\{p_{jt}, p_{jt}^x, p_t^c, W_t, R_{jt}, \mathcal{M}_{jt}\}_{t\geq 0,\, j}$ *such that, taking prices as given, (i) households choose consumption and sectoral investment to maximize lifetime utility subject to their budget constraint; (ii) firms in each sector choose labor, capital, and materials to maximize profits subject to the production technology; (iii) capital evolves according to the law of motion; and (iv) all markets clear. The equilibrium conditions are characterized in Appendix* **??**.

5.3 Equilibrium Properties

As in Herrendorf *et al.* (2021), it is useful for the characterization of the aggregate balanced growth path in the next section to express sectoral investment as a function of *effective* productivity. For convenience all proofs can be found in Appendix A.

Lemma 1. *Investment in sector n at time t is given by*

$$x_{nt} = \mathcal{Z}_{nt}^x \sum_{i=1}^{N} \frac{\chi_{int}}{\exp(\tilde{z}_{it})^{\iota_n}},$$

where effective investment productivity is

$$\mathcal{Z}_{nt}^x \equiv \exp(z_{nt}^x)\left(\sum_{i=1}^{N}\tilde{\omega}_{in}\exp(\tilde{z}_{it})^{\iota_n - 1}\right)^{\frac{1}{\iota_n - 1}},$$

with $\exp(\tilde{z}_{jt}) \equiv \frac{\exp(z_{jt})}{\mathcal{C}_{jt}}$*, and unit cost of production* $\mathcal{C}_{jt} \equiv \left(\mathcal{R}_{jt}^{\alpha_j}\mathcal{W}_t^{1-\alpha_j}\right)^{\gamma_j}\mathcal{M}_{jt}^{1-\gamma_j}$.

Effective productivity in investment sector n, $\mathcal{Z}_{nt}^x$, combines investment-specific techno-

logical change, $\exp(z^x_{nt})$, with a CES aggregator of adjusted sectoral productivities. Because rental rates of capital services and intermediate input prices are heterogeneous across sectors, effective productivity depends on adjusted sectoral productivities defined as $\exp(\tilde{z}_{jt}) \equiv \frac{\exp(z_{jt})}{\mathcal{C}_{jt}}$, where the unit cost of production in sector j is $\mathcal{C}_{jt}$. The values $\mathcal{R}_{jt}$ and $\mathcal{M}_{jt}$ are sector-specific costs of capital services and intermediate inputs, respectively, and $\mathcal{W}_t$ is the cost of labor. Both adjusted productivity and effective investment productivity are endogenous, since sectoral unit costs are determined in equilibrium.

We can then characterize aggregate GDP in the economy as a function of wages.

Lemma 2. *Aggregate value added is proportional to wages,*

$$v_t = \frac{\mathcal{W}_t}{(1-\alpha_n)\gamma_n} \sum_{i=1}^{N} \frac{(1-\alpha_n)\gamma_n}{(1-\alpha_i)\gamma_i} l_{it}$$

where investment in sector n is the numeraire.

Hence, if output labor elasticities are identical across sectors, the unitary labor supply implies that aggregate value added moves one-to-one with the wage in the economy.

5.4 Aggregate Growth Path

Definition *An aggregate growth path (AGP) is an allocation such that aggregate output, consumption, and wages growth at a constant rate, and where the economy's interest rate is constant.*

Such an allocation poses no-restrictions on the sectoral allocation of resources, including capital and investment. When capital-labor ratios are not equalized, as it is the case in our economy, sectoral employment reallocation that is consistent with structural change prevents a constant interest rate. In other words, an aggregate growth path only exists in the limit, as in Acemoglu and Guerrieri (2008). Assume sectoral and investment-specific productivity grows at constant (possibly heterogeneous) rates.

Proposition 1. *Assume that (i) inputs into the production of sectoral investment satisfy $\iota_n \neq 1$ for all n, (ii) sectoral and investment-specific productivity grows at constant (possibly heterogeneous) rates. Then there exists an asymptotic AGP where aggregate GDP, wages, and sectoral capital grow at a constant rate. The growth rate of wages follows:*

$$g_w = \underbrace{\frac{(\Psi g_z)_n}{(\Psi\Gamma(I-\alpha)\mathbf{1})_n}}_{\text{factor neutral}} + \underbrace{\frac{g_{z^x_n} - (\Psi\Gamma\alpha(g_{z^x_n}\mathbf{1} - g_{z^x}))_n}{(\Psi\Gamma(I-\alpha)\mathbf{1})_n}}_{\text{ISTC}}, \tag{7}$$

where $\Psi \equiv (I-(I-\Gamma)M')^{-1}$ is the Leontief inverse. Along the AGP, equilibrium growth rates of

prices are:

$$g_{p_j^x} = g_{z_n^x} - g_{z_j^x} \tag{8}$$

$$g_{p_j} = a_j + b_j g_w \tag{9}$$

$$g_{R_j} = g_{z_n^x} - g_{z_j^x} \tag{10}$$

where $a = \Psi[-g_z + \Gamma\alpha(g_{z_n^x}\mathbf{1} - g_{z^x})]$ *and* $b = \Psi\Gamma(I-\alpha)\mathbf{1}$.

The growth rate of sectoral capital is:

$$g_{k_j} = g_w - g_{z_n^x} + g_{z_j^x} \qquad \forall\, j, \tag{11}$$

so sectoral capital grows at the common wage rate adjusted for the differential between sector j's ISTC and the numeraire sector's ISTC.

Given the elasticity of substitution in the investment aggregator, investment is overtaken either by the sector whose price grows fastest $\iota_n > 1$*, or the sector whose prices grows lowest* $\iota_n < 1$*. The identify of the sector as a function of primitives is determined by the following conditions:*

- ***Gross substitutes*** *(*$\iota_n > 1$*): sector n satisfies:*

$$\frac{g_{z_n^x} - a_n}{b_n} \geq \frac{g_{z_n^x} - a_j}{b_j} \qquad \forall\, j \neq n. \tag{SC-GS}$$

- ***Gross complements*** *(*$\iota_n < 1$*): sector n satisfies:*

$$\frac{g_{z_n^x} - a_n}{b_n} \leq \frac{g_{z_n^x} - a_j}{b_j} \qquad \forall\, j \neq n. \tag{SC-GC}$$

where a_j *and* b_j *are the j-th entries of* $a = \Psi[-g_z + \Gamma\alpha(g_{z_n^x}\mathbf{1} - g_{z^x})]$ *and* $b = \Psi\Gamma(I-\alpha)\mathbf{1}$ *respectively. The growth rate of aggregate value added satisfies* $g_v = g_w = g_{k_n}$*, and the capital in the numeraire sector is pin down from the growth rate of wages.*

Hence, sectoral capital grows proportional to wages and the differential in ISTC. Sectors with faster (Slower) ISTC than the numeraire grow faster (slower) than wages, and the cost of investment decreases (increases) relatively more. Absent ISTC, capital in all sectors grow proportional to wages.

We use this result to detrend the economy and characterize equilibrium allocations.

5.5 Equilibrium Characterization, Detrended Economy

Domar weights. Let the Domar weight of sector n be $\tilde{\eta}_n \equiv \frac{p_n y_n}{p v}$, the share of value added allocated to the production of final goods be $\tilde{\zeta}_n \equiv \frac{p_n c_n}{p v}$ and the value-added share of each sector be $\zeta_n \equiv \tilde{\zeta}_n + \frac{p_n \chi_n^d}{p v}$. We also define adjusted sectoral depreciation rate in the detrended economy $\hat{\delta}_i \equiv 1 - \frac{1-\delta}{1+g_{k_i}}$, the (diagonal) matrixes of adjusted depreciation $\hat{\boldsymbol{\delta}} = \text{diag}\{\hat{\delta}_i\}$, value-added shares $\boldsymbol{\Gamma} = \text{diag}\{\gamma_n\}$, sectoral capital expenditure shares in value added, $\boldsymbol{\alpha} = \text{diag}\{\alpha_n\}$, imported capital expenditure shares, $\boldsymbol{\phi} = \text{diag}\{\phi_n\}$ and the discount factor $\tilde{\boldsymbol{\beta}} = \text{diag}\{\hat{\beta}_i\}$ for $\tilde{\beta}_i \equiv \frac{1}{R} - (1 - \hat{\delta}_{it}) \frac{p_{it+1}^x}{p_{it}^x}$. In what follows, I corresponds to the identity matrix.

Proposition 2. *The equilibrium Domar weights satisfy*

$$\tilde{\boldsymbol{\eta}}_t = \left[I - \tilde{\boldsymbol{\beta}}^{-1} \boldsymbol{k}_{t+1}^{-1} \boldsymbol{x}_t \boldsymbol{\Gamma} \boldsymbol{\alpha} (I - \boldsymbol{\phi}) \boldsymbol{\Omega}_t - (I - \boldsymbol{\Gamma}) \boldsymbol{M} \right]^{-1} \tilde{\boldsymbol{\zeta}}_t \tag{12}$$

In vector form,

$$\tilde{\eta}_{nt} = \tilde{\zeta}_{nt} + \sum_{i=1}^{N} \frac{x_{it}/k_{it+1}}{\tilde{\beta}_i} \alpha_i \gamma_i \omega_{ni} (1 - \phi_i) \tilde{\eta}_{it} + \sum_{i=1}^{N} (1 - \gamma_i) \mu_{ni} \tilde{\eta}_{it}.$$

Proofs to all propositions can be found in Appendix A.2.

Note that the role of the investment network for the Domar weight scales with the importance of domestic investment across sectors, $(1 - \phi_n) \in (0, 1)$.[19] In our economy, equilibrium Domar weights are nontrivial functions of the investment rates along the transition to the AGP, and thus, Domar weights are functions of the entire path of future Domar weights.

Welfare. We start by describing how aggregate welfare, W, in the economy depends on the investment network.

Proposition 3. *Along the equilibrium path, welfare satisfies*

$$W = \frac{\nu}{C} \tilde{\boldsymbol{\eta}} \Gamma \boldsymbol{z},$$

where $\tilde{\boldsymbol{\eta}}$ are the equilibrium Domar weights.[20]

The proof of this result follows from the envelope theorem and is analogous to the extensive literature on production networks studying the implications of roundabout effects on the

[19] This is driven by our modeling of exports from a composite of final uses. If all equipment is tradable, then roundabout effects from the investment network will appear directly in GDP through trade amplification; see Φ in Proposition 4. If instead we model each sector's allocation of gross output to exports, these flows appear in the equilibrium representation of the Domar weight.

[20] The term Γ rescales the productivity vector TFP in the production technology is scaled by value-added share of gross output.

aggregate economy. Value-added shares scale productivity levels because of productivity is scaled by the value added share when defining the technology for sectoral production. Proposition 3 is consistent with results in vom Lehn and Winberry (2022) for short-run fluctuations: Domar weights are scaled by the ratio between GDP and final consumption. Along the AGP, this ratio is a constant. In the transition, the welfare effect of sectoral productivity shocks can be amplified or dampened depending on the relative allocation of value added between consumption and investment uses.

Aggregate value added

Proposition 4. *The equilibrium level of GDP in the economy satisfies*

$$\ln(\nu_t) = \Phi_t \boldsymbol{\eta_t}' \Gamma (z_t + \alpha \boldsymbol{\phi} \Omega_t' \tau_t) + \boldsymbol{\epsilon}_t,$$

where $\Phi_t \boldsymbol{\eta_t} \Gamma$ *is the vector of sectoral influence;* $\Phi_t \equiv (I - \boldsymbol{\eta_t} \Gamma \alpha \boldsymbol{\phi}' \boldsymbol{\Omega_t'})^{-1}$ *is an adjustment factor for the tradable nature of investment; and* $\boldsymbol{\epsilon}_t$ *is an adjustment factor that depends on the equilibrium Domar weights.*[21] *sectoral influence is the product of sectoral value-added shares,* $\boldsymbol{\zeta_t'}$*, and an adjusted Leontief inverse* $\Xi_t \equiv (I - \Gamma\alpha(I - \phi)\Omega_t - (I - \Gamma)M)^{-1}$*, i.e.,* $\boldsymbol{\eta_t} \equiv \boldsymbol{\zeta_t'} \Xi_t$*. In vector form, GDP can be described as*

$$\begin{aligned} \ln(\nu_t)(1 - \sum_n \eta_{nt}\gamma_n\alpha_n \sum_i \omega_{int}\phi_i) \quad &= \quad \sum_n \eta_{nt}\gamma_n z_n + \sum_n \eta_{nt}\gamma_n\alpha_n \sum_i \omega_{int}\phi_i \ln(\tau_t) - \\ &\quad \ln(\sum_n \gamma_n(1-\alpha_n)\tilde{\eta}_{nt}) \sum_n \eta_{nt}\gamma_n(1-\alpha_n). \end{aligned}$$

Value added is thus a function of sectoral TFP, z, the terms of trade, $\boldsymbol{\tau}$, and a residual term $\boldsymbol{\epsilon}$. The first term captures the effect of productivity on value added, operating through the vector of sectoral influence $\Phi_t \boldsymbol{\eta_t} \Gamma$, in the spirit of Acemoglu *et al.* (2012).[22] This influence vector is augmented by $\Phi_t \equiv (I - \boldsymbol{\eta_t} \Gamma \alpha \boldsymbol{\phi}' \boldsymbol{\Omega_t'})^{-1}$, which captures the amplification arising from trade in investment goods, as in Jones (2013). A rise in domestic productivity expands export capacity, and trade balance requires a corresponding increase in imports; when those imports take the form of investment equipment, higher productivity feeds back into capital accumulation. The strength of this channel is increasing in the economy's dependence on imported equipment, ϕ, and in the intensity of capital use in gross output, $\Gamma\alpha$.

The terms of trade enter as a second, distinct channel. Once adjusted for imported investment, the capital share, and the investment network, a change in the terms of trade affects

[21] The term $\boldsymbol{\epsilon}_t \equiv -\Phi \boldsymbol{\eta_t}' \Gamma(1-\alpha) \ln(\Gamma(1-\alpha)\tilde{\boldsymbol{\eta}}_t)$ maps onto the equilibrium distribution of employment and is quantitatively small, so for most of the analysis, it can be omitted.

[22] Value-added shares also mediate this effect because productivity enters the value-added expression directly. If the model is cast in terms of gross output, the factor Γ drops out.

the economy in a manner analogous to a TFP shock. As $\phi \to 0$, the economy's dependence on tradable investment vanishes, and Propositions 2 and 4 collapse to their closed-economy counterparts, which we derive in the Online Appendix.

6 Income Accounting

The model economy that we propose has two key outcomes that help us understand the role of the investment network for the proximate sources of income disparities across countries. The structural relationship between sectorial productivity and aggregate GDP allows us to (a) isolate the effect of changes in sectorial productivity for aggregate GDP; and (b) study how these changes interact with the roundabout nature of the investment and intermediate input linkages to generate aggregate disparities in outcomes. Since the investment network changes endogenously with sectorial productivity trends (and therefore the relative cost of investment in different sectors), we assess the role of the network in general equilibrium in Section 5.5. Before moving there, it is still useful to compare the predictions of our accounting framework for the role of productivity and measured disparities in capital for income disparities.

Recall from Proposition 4 that aggregate value added is

$$\text{GDP} \;=\; \boldsymbol{\eta}\, \boldsymbol{a}, \tag{13}$$

the inner product of sectoral influence and the vector of sectoral "productivity-like" shifters, $\boldsymbol{a} \equiv \boldsymbol{z} + \boldsymbol{\alpha}\, \boldsymbol{\phi}\, \boldsymbol{\Omega}' \boldsymbol{\tau}$, which depend on both sectoral TFP and terms of trade. As in the model economy, terms of trade operate like a sectoral TFP shock. Both objects can be inferred residually using the model's structural restrictions,

$$\boldsymbol{a} \;=\; (\boldsymbol{\Phi \Xi \Gamma}\,)^{-1} \ln(\boldsymbol{\nu}), \tag{14}$$

where $\ln(\boldsymbol{\nu})$ is the vector of logs of sectoral real value added.[23] We use (14) to infer relative productivities across sectors and then pin down the level of productivity to match each country's observed income per capita and the relative sectorial value added shares.

To ensure cross-country comparability, the accounting is conducted for 2005—the only year with PPP sectoral prices we use to convert nominal sectoral value added into real units in the WIOD sample. For countries not in WIOD, we use GDP PPP price deflators from the Penn World Table (PWT). We draw estimates of the capital shares from PWT. A full description of

[23] Separating TFP and the terms-of-trade would require constructing equipment-specific terms of trade.

Table 2: Development Accounting

	Income variance	Contribution of Ω (%)
Baseline	1.00	—
Investment network $\equiv$ VA shares	0.85	15
Channels		
Only investment network	0.37	37
Only investment network + $\boldsymbol{a}_{US}$	0.22	22
Only intermediate-input network	0.82	18
Trade		
Domestic investment network	0.93	7

Notes: Baseline is normalized to 1 and refers to the income variance when GDP per capita is given by Equation (13), matching empirical observations. *Only investment network* excludes intermediate-input linkages; *Only intermediate-input links* excludes investment linkages; *Only investment network* + $\boldsymbol{a}_{US}$ excludes intermediate-input linkages, and fixes sectoral productivity to US;*Investment network* $\equiv$ *VA shares* excludes intermediate-input linkages, and replaces columns of investment network with sectoral value added shares; *Domestic investment network* includes only domestic investment linkages and excludes intermediate-input linkages.

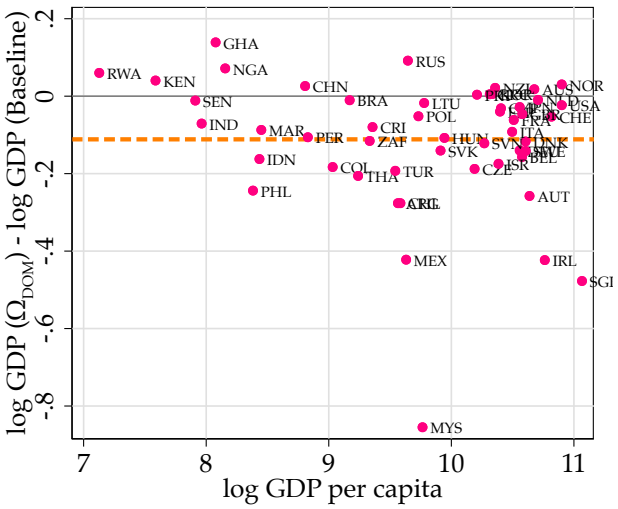
data sources can be found in Appendix B.2.

We assess the role of the investment network for income disparities guided by the insights of our simple economy. In particular, we compare predicted income disparities of an economy where the investment network is replaced by a matrix whose columns correspond to sectorial aggregate value added shares, and therefore all identical. Under this counterfactual networks, we should recover relative contributions of TFP and measured capital output ratios that are comparable to those obtained from standard accounting. We interpret the difference between the income dispersion implied by this economy and the counterfactual, as the role of investment networks in generating income disparities. Since the role of the network is mediated by sectorial productivity, we can decompose the direct effect of disparities in the augmented Leontief-inverse, i.e. the roundabout effects, by imposing the sectorial productivities equal to those in the US. Finally, we study the role of tradable equipment by considering the closed-economy counterparts to the investment network and sectorial influence.

When we restrict the network to have identical columns equal to each sector's value-added share, we reproduce 85% of the income variation in the baseline. The remaining 15% is therefore attributable to having a distinct technology for investment production. The 85% itself reflects the combined contributions of sectoral TFP and capital-output ratios, as in the simple economy of Section 2.

To gauge the magnitude of this additional 15%, we benchmark it against the role of measured capital-output ratio disparities using a standard income-accounting exercise (Jones, 2016); full details are provided in the Online Appendix. Variation in measured capital-output ratios accounts for 13% of income disparities in our sample. Incorporating the investment network

Figure 6: Counterfactual: Domestic Investment Network



Notes: Income per capita computed using domestic investment network for each country, relative to the baseline.

therefore at least doubles the contribution of capital to income accounting, from 13% to 28%.

To assess how these results compare with the broader literature on roundabout effects, we also consider the role of intermediate inputs alone, following Fadinger *et al.* (2022). Excluding the investment network, the model accounts for 82% of the variation in the data, while the investment network in isolation accounts for 37%. These figures do not sum to 100% due to model nonlinearities. Taking the average contribution across both exercises in the spirit of a Shapley decomposition, the investment network accounts for approximately 28% of income variation.

These contributions, however, conflate two distinct forces: the structure of the network itself and the pattern of sectoral productivity disparities across countries. When we impose U.S. sectoral productivities on all countries, the contribution of the investment network falls from 37% to 22%. Roughly one third of its total role reflects the fact that poor countries tend to have low productivity precisely in the sectors whose output is used intensively as investment.

Finally, we quantify the role of capital trade. We do so by shutting down the trade amplification term, Φ, and recomputing the investment network using only *domestic* sectoral investment flows. Average GDP per capita declines by 9% in our sample (Figure 6), which reflects the substantial costs of closing these economies. This decline is larger in high-income economies than in low-income economies because richer economies are most open to trade, see Appendix Figure 19) for a sector-specific measure of import penetration across the development spectrum. Hence, the closed economy version of our economy explains 93% of the observed variation in income-per capita, which implies that 7% can be attributed to trade.

6.1 Sectoral influence

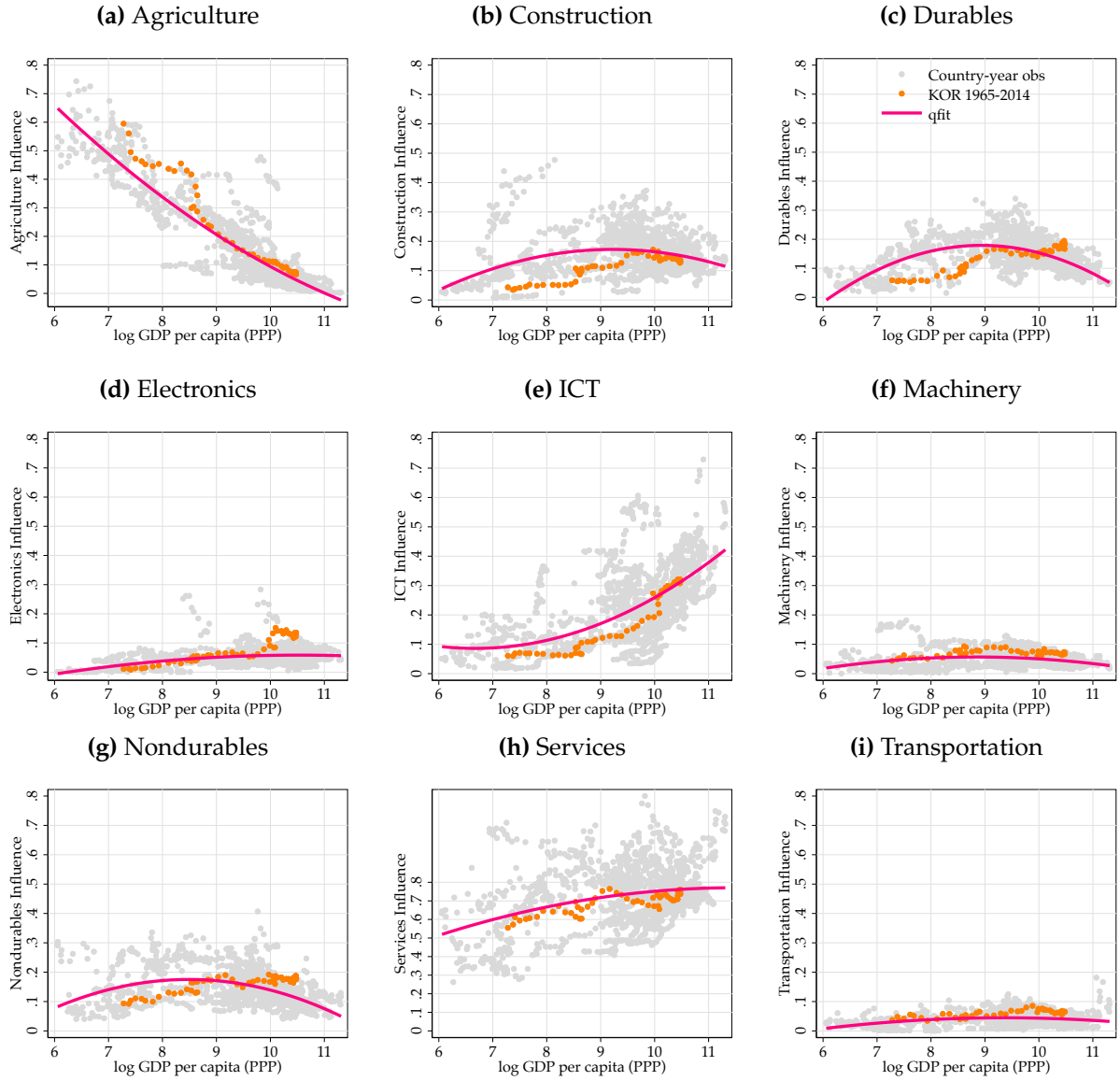
Before moving to general equilibrium, it is useful to assess the differential role of sectors in driving income differences as summarized by sectoral influence, Proposition 4.

$$\tilde{\eta}^{\text{GDP}} \equiv \underbrace{\Phi}_{\substack{\text{trade} \\ \text{amplification}}} \quad \underbrace{\tilde{\zeta}'}_{\text{exp share in VA}} \quad \underbrace{(I - \Gamma\alpha(I-\phi)\Omega - (I-\Gamma)M)^{-1}\Gamma}_{\substack{\text{augmented} \\ \text{Leontief inverse } \Xi}}, \tag{15}$$

The influence vector characterizes how changes in productivity in a given sector contribute to aggregate value added. It is disciplined by the sector's direct contribution to value added and its indirect contribution through investment and intermediate-input linkages, and it is amplified by trade via the imported fraction of investment goods. Figure 7 shows sectoral influence for country-year observations using our full sample. We highlight the development path of South Korea from 1965 to 2014, which aligns surprisingly well with the fitted average across the sample.

Some empirical patterns emerge. A 1% increase in the productivity of the construction sector translates into a 0.13% increase in GDP among low income countries. This influence almost doubles among middle income countries. Sectoral influence for durable goods follows a similar trajectory and magnitude as that of the construction sector despite its lesser role as a producer of investment to other sectors. Sectoral influence for electronics, machinery and transportation is smaller in magnitude along the development spectrum.

Figure 7: Sectoral Influence Along Development



Notes: Data for all country-years in the sample. sectoral influence is defined as $\Phi\eta\Gamma$ in Proposition 4, and measures the change in aggregate value added following a change in sectoral productivity. Dots represent country–year observations; South Korea (1965–2014) is highlighted.

In some sectors, the importance of productivity gains changes substantially as countries develop. The influence of agriculture declines sharply with development, from 0.3% on average in low-income countries to 0.04% in high-income countries. Moreover, a 1% productivity improvement in ICT is linked to a 0.12% increase in GDP per capita in low-income countries, but a much larger 0.34% increase in high-income countries. This likely reflects complementarities and linkages with the rest of the economy, which we explore further in out equilibrium counterfactuals. The influence of other (non-ICT) services deserves special mention, as they exhibit the highest level of influence across countries at all stages of development. Their contribution to aggregate GDP rises with development—from 0.6% to 0.7%—despite a reduction in

their role as providers of investment inputs as countries become richer, documented in Table 1.

One concern with pooling country–year observations across income levels is that the technologies available fifty years ago differ from those available to countries catching up more recently. Our new data also allow further exploration of how investment paths vary across development trajectories. As a first step, in Figure 18 we highlight the paths of India and China, two countries that have experienced relative rapid growth since 2000s, when the technologies available for production were arguably different than those available in the 1960s. Even for these cases, the path of sectoral influence—especially for newer technologies such as ICT—tracks the pattern predicted by their income levels.[24]

Another concern is that these patterns could be driven entirely by standard structural-transformation forces —namely, the reallocation of sectoral value added (ζ) as countries develop. Since value added shares and the investment network change in equilibrium with the path of sectorial productivity, this decomposition is better studied in equilibrium, which we do next.

7 The role of the investment network in general equilibrium

We study the transition dynamics of the model economy calibrated to South Korea over the period 1965–2014. The goal is to quantify the contribution of relative price changes to the evolution of the investment network and sectoral influence along the development path, and to assess the role of productivity improvements in different sectors for aggregate output and (measured) capital-output ratios. To ease the computation, we solve an economy with five sectors: Agriculture, Manufacturing, Construction, ICT and Services.

Table 3 summarizes the parameter values. We discipline the investment aggregator in each sector through the elasticity of substitution, ι_s, and the CES loadings $\tilde{\omega}_{in}$ by asking our model to replicate key coefficients of a panel regression of log relative investment expenditure shares on log relative prices, using South Korean data from 1965 to 2014. The input–output network, $\mu_{in,t}$, is interpolated linearly between its 1965 and 2014 values. Sectoral TFP growth rates are chosen to match the changes in relative prices in this time horizon while its growth rate in the manufacturing sector (the base sector) disciplines the levels of aggregate GDP per capita change observed in the data. The levels of TFP are chosen so that value added shares in 2014 match those in the data. We also discipline the degree of income effects in preferences by targeting consumption shares, for a given elasticity of substitution in consumption, set as in

[24] One exception is other services, where sectoral influence lies below the average path for China and above it for India.

Buera, Kaboski, Mestieri and O'Connor (2020).

The economy starts at the balanced growth path consistent with initial TFP levels and the 1965 input–output structure, and transitions as TFP grows linearly over 45 periods (1965–2010), after which TFP tapers smoothly to its terminal level.

Table 3: Calibration: Korea Transition Dynamics

Parameter	Description	Value	Moment/Calibration
		Common	
Technology			
γ	VA share in gross output	0.50	Intermediate input share in gross output
α	Capital share in VA	0.33	Standard
δ	Depreciation rate	0.10	Standard
ι	Elast. of subst., inv.	1.20	Regression investment shares on prices
$\tilde{\omega}_{in,t}$	Investment weights	see Table 16	Sectorial investment expenditure shares
$\mu_{in,t}$	IO network	see Table 17	Input-output network
Preferences			
β	Discount factor	0.96	4% interest rate
σ	Elast. of subst., cons	0.0001	Buera *et al.* (2020)
Euler wedge			
τ_t	Wedge in Euler eq.	See Figure XX	Aggregate investment ratio $I/(C+I)$
		Sectorial	
Preferences			
ϵ_j	Income elasticities	$\{0.05, 1.00, 1.20, 1.00, 1.20\}$	Comin *et al.* (2021), external
θ_j	Consumption weights	$\{0.07, 0.00, 0.07, 0.29, 0.57\}$	Average consumption shares, 1965–2014
TFP			
$z_{j,2014}$	Sectoral log TFP at 2014	see Figure **??**	VA shares 2014
Δz_j	Sectoral log TFP growth	see Figure **??**	Relative prices, 1965–2014
$\Delta \bar{z}$	Aggregate log TFP growth	1.67	GDP per capita ratio 2014/1965

Notes: Common and sector-specific parameters in the baseline 5-sector calibration. Sector ordering is Agriculture, Construction, ICT, Manufacturing, Services. $\tilde{\omega}_{in,t}$ and $\mu_{in,t}$ are time-varying and reported at the three anchor years 1965, 1990, 2014 in Appendix Tables 16 and 17; both are interpolated between anchors and held flat after 2014. The Euler wedge τ_t is recovered in closed form from the BGP identity so that the model's aggregate investment share $I/(C+I)$ matches the data at each year of matching; values reported as $\{\tau_{1965}, \tau_{1990}, \tau_{2014}\}$.

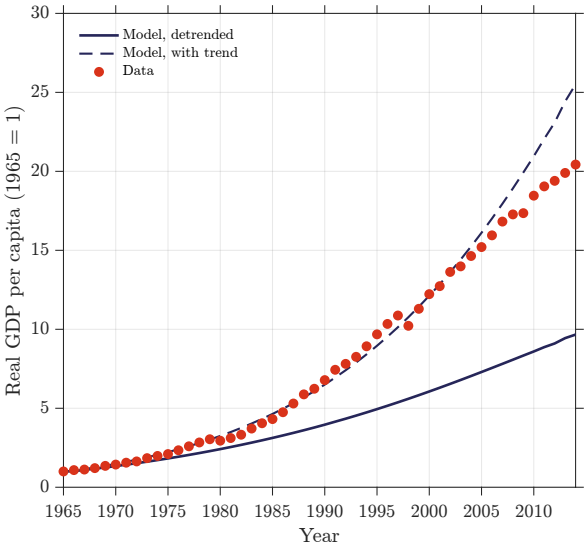
Targeted Moments

The degree of income effects, sectoral TFP levels and growth rates, and the Euler-equation wedge τ_t are calibrated to jointly match the trends in relative sectorial prices, the level of GDP per capita growth, the aggregate investment share of domestic absorption $I/(C+I)$, and the average level of value-added shares in the sample. Figure 8 compares the model with the data along these three targeted dimensions: GDP per capita (panel a), the investment share of domestic absorption (panel b), and relative prices (panel c). The model tracks the path of relative prices well. One salient feature of calibrating our economy to the path of Korea rather than other industrialized countries is that the relative price of agriculture relative to manufacturing is rising over the sample period, rather than declining as it is commonly featured for the process

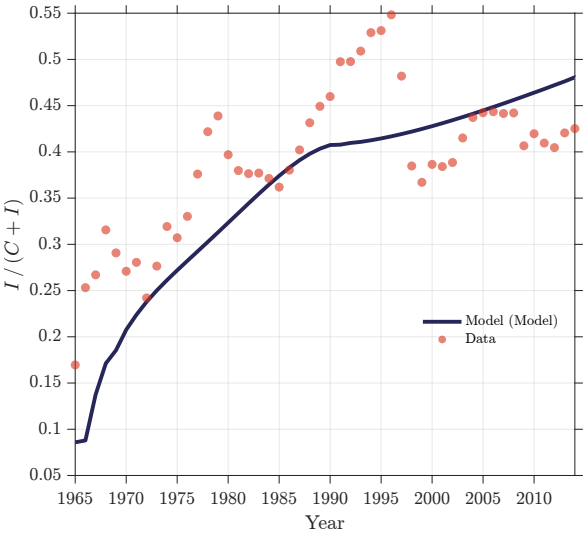
of structural change, Herrendorf, Rogerson and Ákos Valentinyi (2014). The model rationalizes the rise in the relative price with faster productivity growth in manufacturing. The investment share of absorption rises from about 17% in 1965 to over 40% by the late 1990s, which the model matches through the calibrated wedge τ_t.

Figure 8: Targeted Moments: Model vs Data (South Korea, 1965–2014)

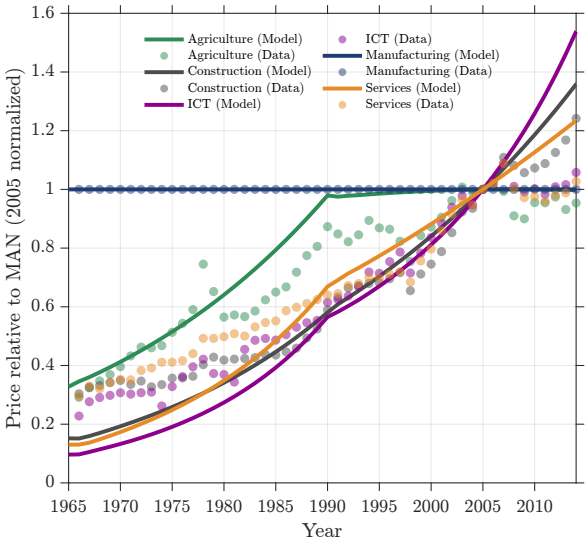
(a) GDP per capita (1965 = 1)



(b) Investment Share, $I/(C+I)$



(c) Relative Prices (2005 = 1)

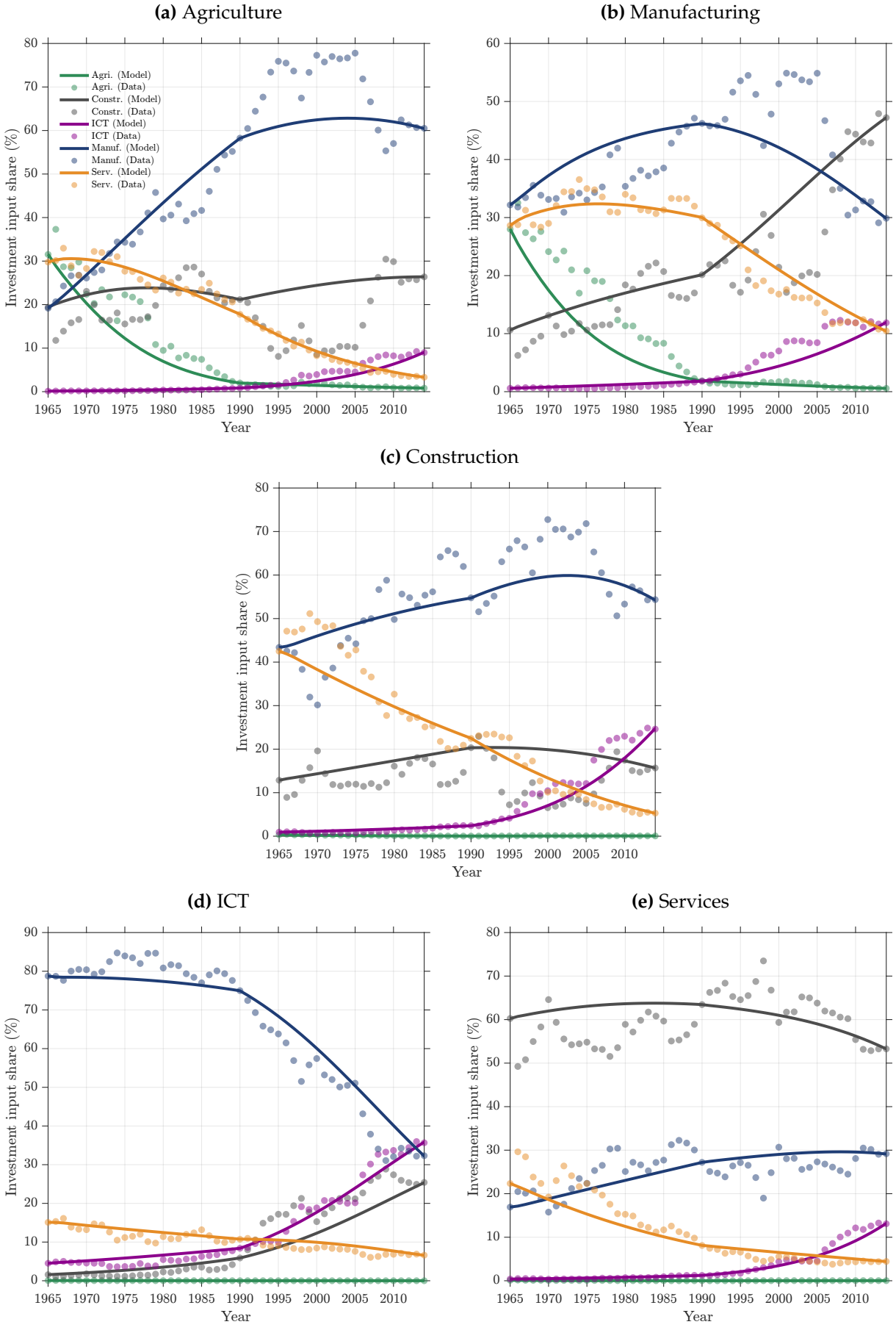


A further targeted dimension of the calibration is the path of investment expenditure shares in each destination sector. By construction, the CES investment loadings $\tilde{\omega}_{in,t}$ are recovered exactly from observed sectoral investment shares, so the model reproduces the data shares almost perfectly (Figure 9). In the data, the share of manufacturing investment goods rises throughout the period in all sectors except ICT, the share of services falls in all sectors, and ICT equipment increases substantially as an investment input in all sectors.[25]

[25] A calibration where the investment loadings are kept constant and the only drivers of changes in the investment

network are relative prices induces most of the qualitative changes in the network, with the exception of the path of expenditure shares in services and in the construction sector.

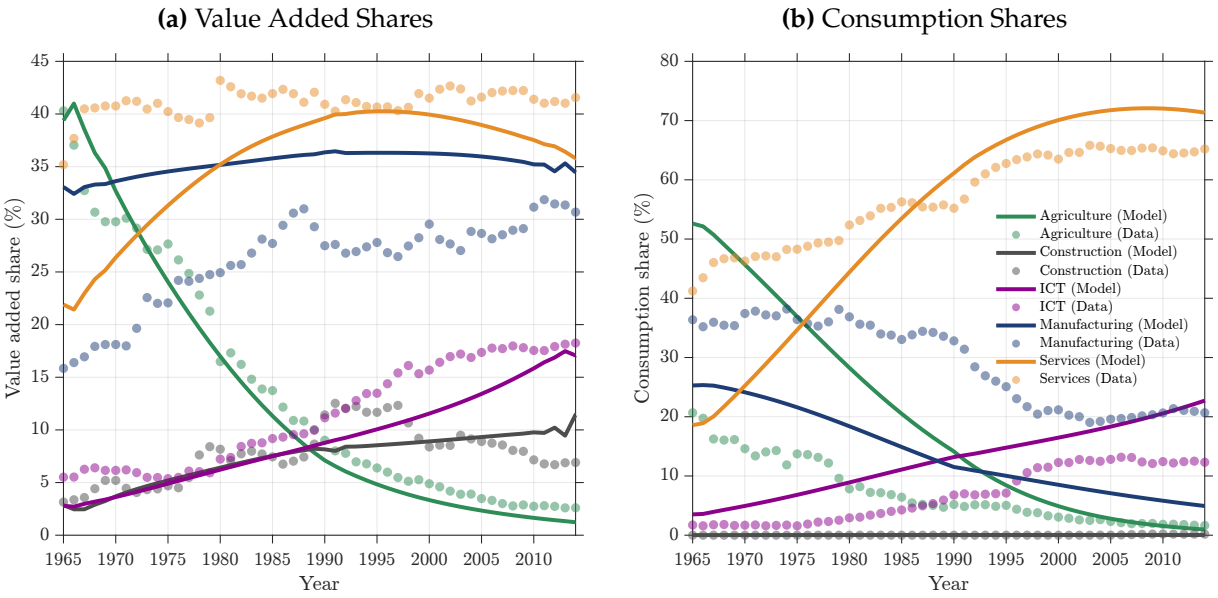
Figure 9: Investment Input Shares by Destination Sector: Model vs Data



Untargeted Moments

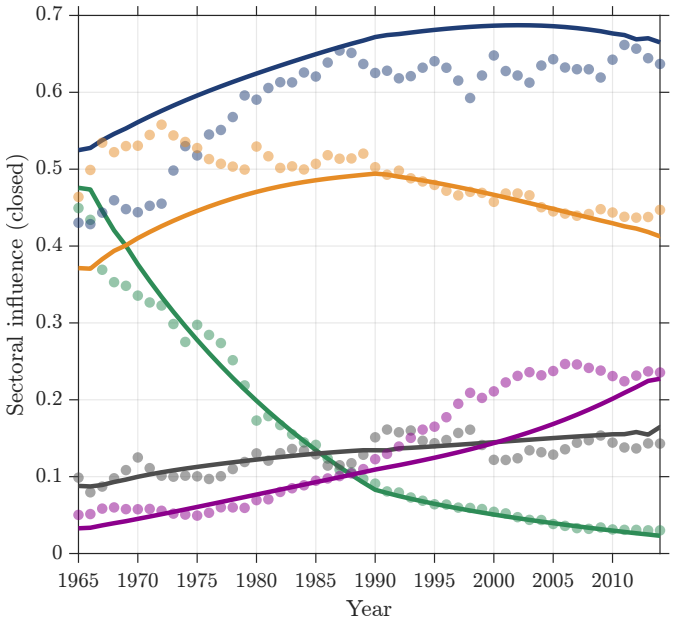
Figure 10 reports the trends in sectoral value added shares and consumption shares, untargeted by our calibration. The model tracks the reallocation of value added from agriculture towards manufacturing as well as services (particularly ICT); in tandem with a value-added share of construction that is relatively stable over the sample period. The model predicts higher increase in services than observed in the data, but matches well the raise in ICT (despite the level, of value added share being slightly lower than in the data). The model captures the increase in ICT and Service expenditure shares from roughly 0% and 40% to 12% and 65% respectively, although it predicts a higher consumption share in agriculture and a lower share in manufacturing than the data. the model underestimates the consumption share of manufacturing, and overestimates is value added share. The path of construction and agriculture value added shares are broadly consistent with the data.

Figure 10: Untargeted Moments: Model vs Data (South Korea, 1965–2014)



Given the estimates of value added shares and the investment network, we can measure sectorial influence. Figure 11 reports sectoral influence using Proposition 4. The model matches the broad patterns well: agricultural influence declines, manufacturing and ICT influence increases, while construction influence is relatively stable. Services influence is predicted to decline slightly with development while in the data service influence increases initially and then declines. The model overestimates agriculture influence at the beginning of the sample. The model is broadly in line with the strong influence of manufacturing and ICT.

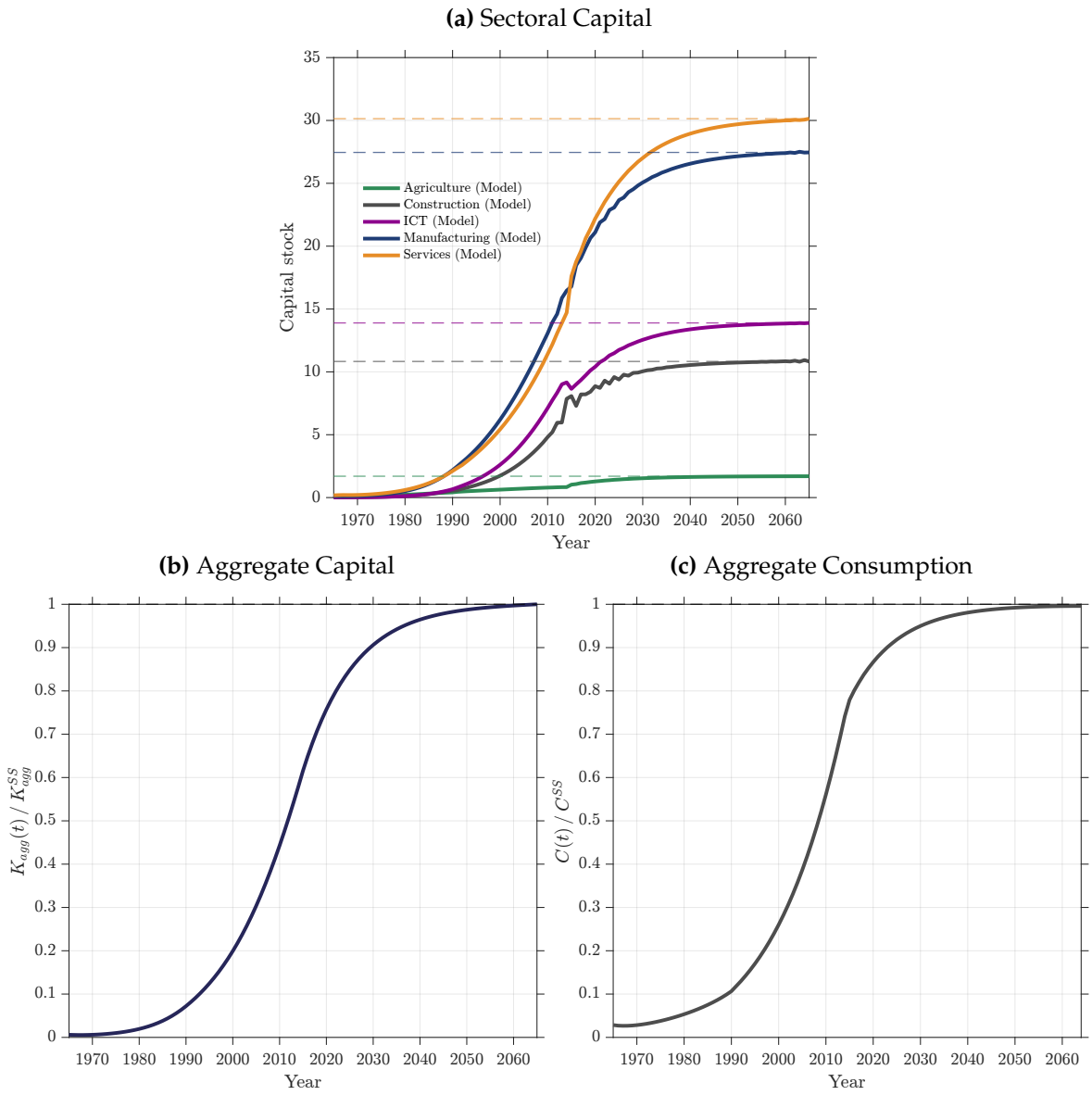
Figure 11: Sectoral Influence: Model vs Data



Transition Paths

Figure 12 reports the transition paths of sectoral capital and aggregate capital and consumption in levels. Capital accumulation is concentrated in manufacturing and services, followed by capital accumulation in the ICT and construction sectors. Agricultural capital grows modestly. Sectoral consumption rises in all sectors, leading to a 7-fold aggregate consumption gain throughout the period. Aggregate capital starts at about 2% of the steady state level and raises close to 68% of its steady-state level by 2014. The economy converges to its steady state by 2065. Aggregate consumption is above 80% of its steady state level by 2014. The catch in investment and consumption reflect standard neoclassical forces behind the transition dynamics of Korea.

Figure 12: Transition Paths: Levels



7.1 Sectorial productivity

With a fully calibrated economy we can now revisit the role of capital accumulation for the transition dynamics. Relative to the income accounting exercises, the equilibrium counterfactuals reflect the endogenous response of value added shares and the investment network to relative prices.

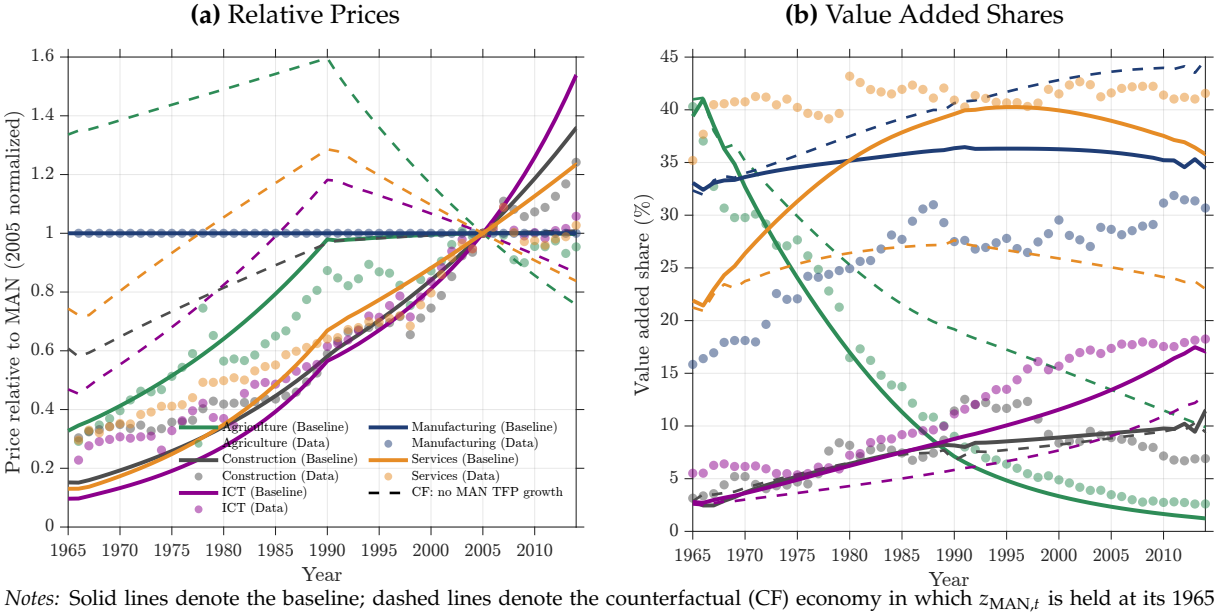
Our main exercises shut the path of productivity growth in different sectors of the economy, to understand its impact on aggregates and sectorial reallocation. We present counterfactual manufacturing productivity path, and we are working on the counterfactuals where we shut productivity growth in ICT and construction.

7.2 The role of manufacturing productivity

We construct a second counterfactual that shuts down TFP growth in the manufacturing sector while keeping every other primitive at its baseline-calibrated value. Specifically, $z_{\text{MAN},t}$ is pinned at its 1965 level for the entire transition, and the remaining sectoral TFP paths, the investment loadings $\tilde{\omega}_{in,t}$, the IO network $\mu_{in,t}$, the consumption weights θ_j and the Euler wedge τ_t are unchanged. Manufacturing is the fastest-growing sector in our baseline calibration, so this experiment quantifies how much of Korea's transition is driven by productivity gains in that single sector.

We start by describing the counterfactual path for relative prices and value added shares, standard inputs to understand the path of structural change. First, despite shutting down productivity growth in manufacturing the prices in all sectors raise in the first half of the sample, up to 1990. The reason for these shifts is that since sectors are heterogeneous in their investment elasticities to sectorial investment, the relative cost of production is not the same in all sectors, which leads to relative price movements beyond those predicted by relative TFP. Second, the slow down in productivity in the manufacturing sector induces a substantial decline in the level of the value added share in services and ICT; as well as an increase in the value added share of agriculture, consistently with a slow-down in the process of structural changes. Perhaps surprisingly, general equilibrium forces induce an increase in the value added share of manufacturing through time to compensate for the slower growth in services.

Figure 13: Counterfactual: No TFP Growth in Manufacturing

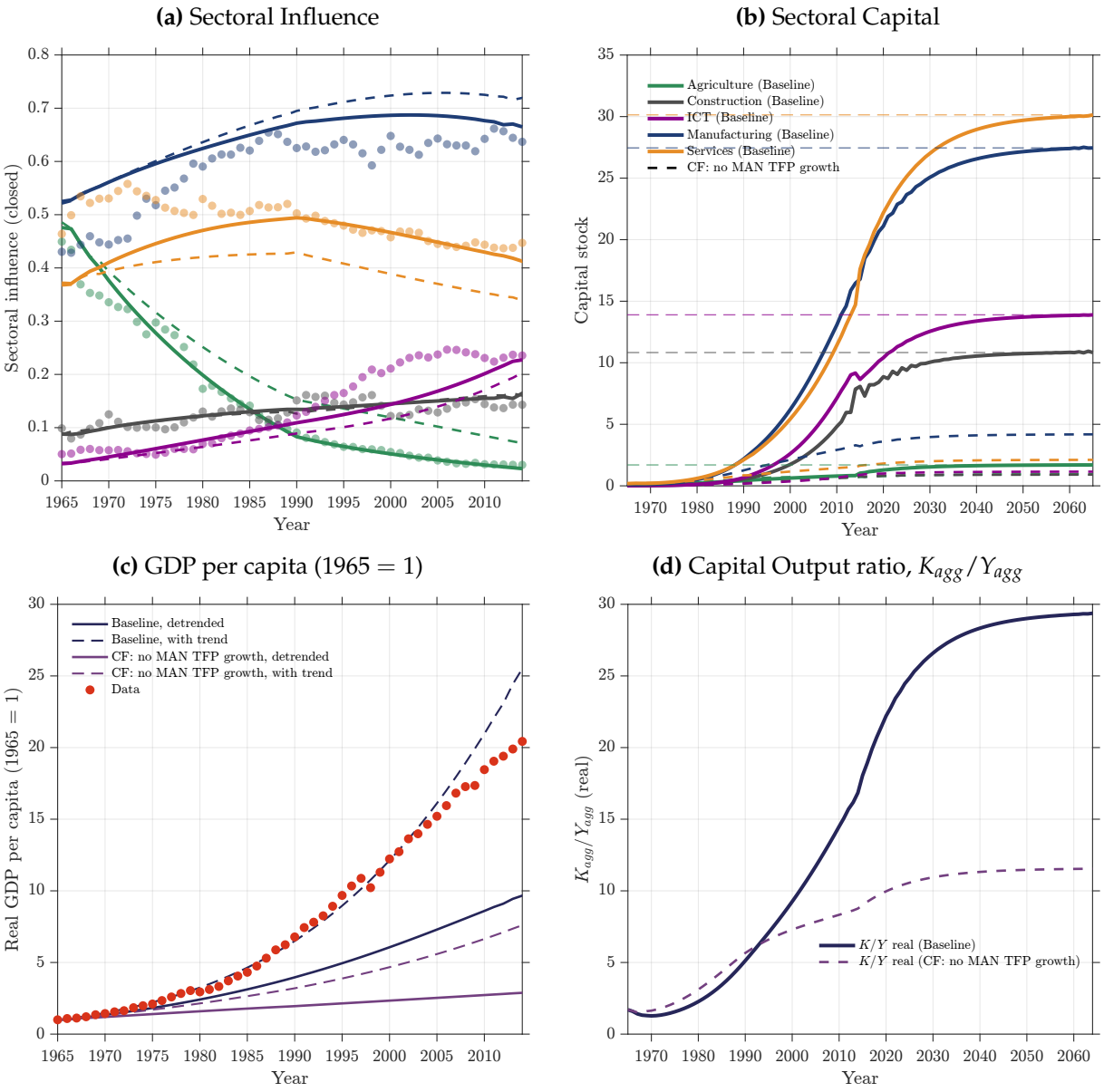


Notes: Solid lines denote the baseline; dashed lines denote the counterfactual (CF) economy in which $z_{\text{MAN},t}$ is held at its 1965 value for the entire transition.

Sectorial influence of agriculture raises and of services falls in the counterfactual, consistent

with standard structural change forces, through value added reallocation. Real GDP growth plummets, and instead of observing more than a 20-fold increase in GDP per capita, the model predicts a 7-fold increase over the period. In the detrended economy, the transition dynamics of the economy implies a 10 fold increase in output along the transition, which gets reduced to a 2.5 fold increase in output absent productivity growth in manufacturing. Such a slow-down in output is associated with a three fold decline in capital output ratios accumulation along the transition.

Figure 14: Counterfactual: No TFP Growth in Manufacturing



Notes: Solid lines denote the baseline; dashed lines denote the counterfactual (CF) economy in which $z_{\text{MAN},t}$ is held at its 1965 value for the entire transition. All other primitives ($\tilde{\omega}_{in,t}$, $\mu_{in,t}$, θ_j, τ_t, $z_{j,t}$ for $j \neq$ MAN, ι, σ, ϵ, β, α, γ, δ) are kept at their baseline-calibrated values. Aggregate capital is reported relative to the *baseline* terminal SS so the level gap between the two paths is visible.

8 Final Remarks

We have constructed novel measures of the investment network for 58 countries across the development spectrum and time-series estimates that cover the period 1965 to 2014. Our analysis reveals systematic disparities in the sectors' roles as providers of investment goods as economies progress. We also document significant empirical disparities between the investment network and the input–output network for countries at different income levels.

Leveraging our estimates of the investment network across countries at different development stages, we conduct an income accounting exercise, finding that disparities in the investment network can account for 28% of observed differences in income per capita across countries, almost double the effect of capital in the standard income accounting exercise.

We argue that this role relates to systematic shifts in the investment network as countries develop. The extent to which comparative advantage, distortions, or variations in human capital endowments explain shifts in the network is an exciting avenue for future research. Is there a systematic ladder in the type of investment required to transition an economy from low to high income levels? We hope that this work can serve as a foundation for studying this and other critical questions.

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A Proofs & Derivations

A.1 Aggregate Growth Path

Proof of Lemma 1. Investment in each sector can be rewritten using optimal expenditure shares (from the problem of the investment firm in each sector),

$$\frac{p_{it}\chi_{int}}{p_{jt}\chi_{jnt}} = \frac{\tilde{\omega}_{in}}{\tilde{\omega}_{jn}}\left(\frac{p_{it}}{p_{jt}}\right)^{1-\iota_n} \tag{16}$$

and the optimality conditions for capital demand at the sectoral level,

$$\frac{p_{it}}{p_{jt}} = \frac{\exp(z_{jt})}{\exp(z_{it})}\frac{\mathcal{C}_{it}}{\mathcal{C}_{jt}} = \frac{\exp(\tilde{z}_{jt})}{\exp(\tilde{z}_{it})} \tag{17}$$

where $\mathcal{C}_{it} \equiv \left(\mathcal{R}_{it}^{\alpha_i}\mathcal{W}_t^{(1-\alpha_i)}\right)^{\gamma_i}\mathcal{M}_{it}^{1-\gamma_i}$ is the marginal cost per unit of output in the sector. The investment aggregator can be written in units of some arbitrary good j as

$$x_{nt} = \exp(z_{nt}^x)\left(\sum_{i=1}^{N}\tilde{\omega}_{in}^{\frac{1}{\iota_n}}\frac{\chi_{int}}{\chi_{jnt}}^{\frac{\iota_n-1}{\iota_n}}\right)^{\frac{\iota_n}{\iota_n-1}}\chi_{jnt}. \tag{18}$$

Replacing the optimality condition in the investment sector and capital demands, equations 16 and 17, we obtain

$$\frac{\chi_{int}}{\chi_{jnt}} = \frac{\tilde{\omega}_{in}}{\tilde{\omega}_{jn}}\left(\frac{\exp(z_{jt})}{\exp(z_{it})}\frac{\mathcal{C}_{it}}{\mathcal{C}_{jt}}\right)^{-\iota_n} = \frac{\tilde{\omega}_{in}}{\tilde{\omega}_{jn}}\left(\frac{\exp(\tilde{z}_{jt})}{\exp(\tilde{z}_{it})}\right)^{-\iota_n}$$

Hence, the investment aggregator reads

$$x_{nt} = \mathcal{Z}_{nt}^x\left(\sum_{i=1}^{N}\tilde{\omega}_{in}^{\frac{1}{\iota_n}}\left(\frac{\tilde{\omega}_{in}}{\tilde{\omega}_{jn}}\left(\frac{\exp(\tilde{z}_{it})}{\exp(\tilde{z}_{jt})}\right)^{\iota_n}\right)^{\frac{\iota_n-1}{\iota_n}}\right)^{\frac{\iota_n}{\iota_n-1}}\chi_{jnt}.$$

which we can rewrite as

$$\tilde{\omega}_{jn}\exp(\tilde{z}_{jt})^{\iota_n-1}x_{nt} = \mathcal{Z}_{nt}^x\left(\sum_{i=1}^{N}\tilde{\omega}_{in}\left(\exp(\tilde{z}_{it})\right)^{\iota_n-1}\right)^{\frac{\iota_n}{\iota_n-1}}\frac{\chi_{jnt}}{\exp\tilde{z}_{jt}}.$$

Adding the above conditions across different j we obtain the equilibrium aggregate investment as a function of cost,

$$x_{nt} = \underbrace{\exp(z_{nt}^x)\left(\sum_{i=1}^{N}\tilde{\omega}_{in}\left(\exp(\tilde{z}_{it})\right)^{\iota_n-1}\right)^{\frac{1}{\iota_n-1}}}_{\mathcal{Z}_{nt}^x}\sum_{i=1}^{N}\frac{\chi_{int}}{\exp\tilde{z}_{it}}. \tag{19}$$

□

Proof. Proof of Lemma 2.

The labor first-order condition in sector i implies

$$\mathcal{W}_t = p_{it}\frac{\partial y_{it}}{\partial l_{it}} = p_{it}(1-\alpha_i)\gamma_i\frac{y_{it}}{l_{it}},$$

so that

$$\mathcal{W}_t l_{it} = (1-\alpha_i)\gamma_i\, p_{it}y_{it}.$$

Nominal value added is defined as

$$VA_{it} \equiv p_{it}y_{it} - \mathcal{M}_{it}m_{it}.$$

Under cost minimization for intermediate inputs, a constant fraction of nominal gross output is spent on materials. In particular, since gross output is Cobb–Douglas in value added and intermediate inputs, the first-order condition for materials implies

$$\mathcal{M}_{it} = p_{it}\frac{\partial y_{it}}{\partial m_{it}} = (1-\gamma_i)p_{it}\frac{y_{it}}{m_{it}},$$

which yields $\mathcal{M}_{it}m_{it} = (1-\gamma_i)p_{it}y_{it}$. Substituting this expression into the definition of value added yields

$$VA_{it} = \gamma_i p_{it}y_{it}.$$

Combining the expressions above yields

$$VA_{it} = \frac{W_t}{1-\alpha_i}\, l_{it}.$$

Aggregating across sectors,

$$VA_t = \sum_{i=1}^{N} VA_{it} = \mathcal{W}_t \sum_{i=1}^{N}\frac{l_{it}}{1-\alpha_i},$$

where both value added and wages are denominated in units of the numeraire.

If capital shares are common across sectors, $\alpha_i = \alpha$, and the labor supply is normalized to one, $\sum_i l_{it} = 1$, aggregate nominal value added is proportional to the wage rate, $VA_t = \frac{\mathcal{W}_t}{1-\alpha}$, independently of the employment allocation in the economy. □

Proof of Proposition 1. **Step 1: Convergence of expenditure shares.**

The expenditure share of sector i in investment for sector j is:

$$s_{ij} = \frac{\tilde{\omega}_{ij} p_{it}^{1-\iota_j}}{\sum_k \tilde{\omega}_{kj} p_{kt}^{1-\iota_j}}. \tag{20}$$

The limiting behavior of these shares depends on the sign of $1 - \iota_j$:

- **Gross substitutes** ($\iota_j > 1$, so $1 - \iota_j < 0$): the sector with the *lowest* p_{it} dominates the sum, so $s_{nj} \to 1$ for the sector n with the fastest declining output price, and $s_{ij} \to 0$ for all $i \neq n$.
- **Gross complements** ($\iota_j < 1$, so $1 - \iota_j > 0$): the sector with the *highest* p_{it} dominates the sum, so $s_{nj} \to 1$ for the sector n with the slowest declining (or fastest rising) output price, and $s_{ij} \to 0$ for all $i \neq n$.

In both cases the expenditure shares converge to a degenerate distribution on a single sector n, and the investment network Ω — through the technological weights $\tilde{\omega}_{ij}$ — becomes irrelevant for the AGP. The only difference across cases is the identity of the dominant sector n.

Step 2: Investment price growth.

In both cases, since $s_{nj} \to 1$, the investment price index collapses to:

$$p_{jt}^x \propto \frac{p_{nt}}{\exp(z_{jt}^x)}, \tag{21}$$

so that:

$$g_{p_j^x} = g_{p_n} - g_{z_j^x}. \tag{22}$$

Step 3: Numeraire and key growth rates.

Take investment in sector n as numeraire, so $p_{nt}^x = 1$ and $g_{p_n^x} = 0$. Since $p_{nt}^x \propto p_{nt} / \exp(z_{nt}^x)$, holding $p_{nt}^x = 1$ requires:

$$g_{p_n} = g_{z_n^x}. \tag{23}$$

It is the *output price* of sector n that grows at rate $g_{z_n^x}$. Substituting (23) into (22):

$$g_{p_j^x} = g_{z_n^x} - g_{z_j^x} \qquad \forall\, j. \tag{8}$$

Note that $g_{p_n^x} = 0$ by construction. Sectors with faster ISTC $g_{z_j^x}$ than $g_{z_n^x}$ see falling investment prices; sectors with slower $g_{z_j^x}$ see rising investment prices. Along the AGP, the Euler equation gives:

$$g_{R_j} = g_{p_j^x} = g_{z_n^x} - g_{z_j^x} \qquad \forall\, j, \tag{10}$$

In the numeraire sector, since $g_{p_n^x} = 0$, then $g_{R_n} = 0$, consistent with a constant interest rate along the AGP.

Step 4: Price system and wage decomposition.

From cost minimization, $p_{jt} = C_{jt}/\exp(z_{jt})$, so $g_{C_j} = g_{p_j} + g_{z_j}$ and effective productivity satisfies $g_{\tilde{z}_j} = g_{z_j} - g_{C_j} = -g_{p_j}$. Substituting $g_{R_j} = g_{z_n^x} - g_{z_j^x}$ into the cost minimization condition:

$$g_{p_j} = -g_{z_j} + \gamma_j\left[\alpha_j(g_{z_n^x} - g_{z_j^x}) + (1-\alpha_j)g_w\right] + (1-\gamma_j)\sum_i \mu_{ij}\, g_{p_i}. \tag{24}$$

In matrix form:

$$(I - (I-\Gamma)M')\, g_p = -g_z + \Gamma\alpha(g_{z_n^x}\mathbf{1} - g_{z^x}) + \Gamma(I-\alpha)\mathbf{1}\, g_w. \tag{25}$$

Define Leontief inverse $\Psi \equiv (I - (I-\Gamma)M')^{-1}$, then:

$$g_p = \underbrace{\Psi[-g_z + \Gamma\alpha(g_{z_n^x}\mathbf{1} - g_{z^x})]}_{a} + \underbrace{\Psi\Gamma(I-\alpha)\mathbf{1}}_{b}\, g_w \equiv a + b\, g_w. \tag{9}$$

Applying $g_{p_n} = g_{z_n^x}$ and selecting row n:

$$g_{z_n^x} = a_n + b_n\, g_w \implies g_w = \frac{g_{z_n^x} - a_n}{b_n}. \tag{26}$$

Substituting $a_n = -(\Psi g_z)_n + (\Psi\Gamma\alpha(g_{z_n^x}\mathbf{1} - g_{z^x}))_n$ and $b_n = (\Psi\Gamma(I-\alpha)\mathbf{1})_n$ delivers the decomposition:

$$g_w = \frac{(\Psi g_z)_n}{(\Psi\Gamma(I-\alpha)\mathbf{1})_n} + \frac{g_{z_n^x} - (\Psi\Gamma\alpha(g_{z_n^x}\mathbf{1} - g_{z^x}))_n}{(\Psi\Gamma(I-\alpha)\mathbf{1})_n}. \tag{7}$$

The factor-neutral term is a Leontief-inverse-weighted average of sectoral TFP growth normalized by labor shares. The ISTC term captures the direct effect of investment-specific productivity in sector n and the indirect effect through differential rental rates across all sectors, weighted by capital shares and the IO structure. Both terms and the price system are identical across the gross substitutes and gross complements cases — only the identity of sector n differs.

Step 5: Consistency of the converging sector.

- **Gross substitutes** ($\iota_j > 1$): sector n has the fastest declining price, so $g_{p_n} \leq g_{p_j}$ for all $j \neq n$. Substituting $g_{p_n} = g_{z_n^x}$ and $g_{p_j} = a_j + b_j g_w$ and rearranging:

$$\boxed{\frac{g_{z_n^x} - a_n}{b_n} \geq \frac{g_{z_n^x} - a_j}{b_j} \qquad \forall\, j \neq n.} \tag{SC-GS}$$

Sector n *maximizes* $(g_{z_n^x} - a_j)/b_j$ across all sectors, consistent with (SC-GS).

- **Gross complements** ($\iota_j < 1$): sector n has the slowest declining (or fastest rising) price, so $g_{p_n} \geq g_{p_j}$ for all $j \neq n$. The same substitution gives:

$$\boxed{\frac{g_{z_n^x} - a_n}{b_n} \leq \frac{g_{z_n^x} - a_j}{b_j} \qquad \forall\, j \neq n.} \tag{SC-GC}$$

Sector n *minimizes* $(g_{z_n^x} - a_j)/b_j$ across all sectors, consistent with (SC-GC).

Step 6: Growth rate of aggregate value added. From Lemma 2, aggregate value added is proportional to the wage. The capital-labor ratios in the numeraire sector, and no labor reallocation along the AGP ($g_{l_n} = 0$) yield:

$$g_v = g_w = g_{k_n}. \tag{27}$$

Since all growth rates are functions of the underlying constant primitive productivity growth rates, they are themselves constant, confirming the existence of the AGP.

Step 7: Growth rate of sectoral capital. From the capital demand condition $R_{jt} = \alpha_j \gamma_j p_{jt} y_{jt}/k_{jt}$, taking growth rates and using $g_{R_j} = g_{p_j^x} = g_{z_n^x} - g_{z_j^x}$:

$$g_{k_j} = g_{p_j} + g_{y_j} - (g_{z_n^x} - g_{z_j^x}). \tag{28}$$

From $W_t l_{jt} = (1 - \alpha_j)\gamma_j p_{jt} y_{jt}$, taking growth rates:

$$g_w + g_{l_j} = g_{p_j} + g_{y_j}. \tag{29}$$

Substituting (29) into (28):

$$g_{k_j} = g_w + g_{l_j} - g_{z_n^x} + g_{z_j^x}. \tag{30}$$

Since $g_{l_j} = 0$ in the asymptotic BGP, the result follows.

Step 8: Growth rate of sectoral consumption. We show that $g_{c_j} = g_{y_j} = g_w - g_{p_j}$ for all j.

Growth of sectoral investment. From the capital accumulation equation $k_{j,t+1} = x_{jt} + (1 - \delta_j)k_{jt}$, along the AGP where k_{jt} grows at the constant rate g_{k_j}, so x_{jt} is proportional to k_{jt} and $g_{x_j} = g_{k_j}$.

Growth of investment inputs. From cost minimization with the CES investment aggregator (6), nominal expenditure on goods from sector j in the production of investment for sector i satisfies $p_{jt}\chi_{jit} = s_{ji,t}\, p_{it}^x\, x_{it}$, where $s_{ji,t}$ is the expenditure share. Along the AGP, expenditure

shares converge to constants (Step 1), so $g_{s_{ji}} = 0$. Taking growth rates and using (8) and (11):

$$g_{p_j} + g_{\chi_{ji}} = g_{p_i^x} + g_{x_i} = (g_{z_n^x} - g_{z_i^x}) + (g_w - g_{z_n^x} + g_{z_i^x}) = g_w, \tag{31}$$

so $g_{\chi_{ji}} = g_w - g_{p_j} = g_{y_j}$ for all i. Since this holds for every destination sector i, the weighted average satisfies $g_{\sum_i \chi_{ji}} = g_{y_j}$.

Growth of intermediate inputs. From cost minimization with the Cobb-Douglas intermediate aggregator, nominal expenditure on goods from sector j used as intermediates in sector i satisfies $p_{jt}\, m_{jit} = \mu_{ji}(1-\gamma_i)\, p_{it}\, y_{it}$. Taking growth rates and using (29):

$$g_{p_j} + g_{m_{ji}} = g_{p_i} + g_{y_i} = g_w, \tag{32}$$

so $g_{m_{ji}} = g_w - g_{p_j} = g_{y_j}$ for all i. Since the growth rate is identical across all destination sectors i, the weighted average satisfies $g_{\sum_i m_{ji}} = g_{y_j}$.

Goods market clearing. Since $g_{\sum_i \chi_{ji}} = g_{\sum_i m_{ji}} = g_{y_j}$, all components of the goods market clearing condition $y_{jt} = c_{jt} + \sum_i \chi_{jit} + \sum_i m_{jit}$ grow at the same rate, so:

$$g_{c_j} = g_{y_j} = g_w - g_{p_j} \qquad \forall\, j. \tag{33}$$

□

Corollary 1 (BGP without Investment-Specific Technical Change)**.** *Suppose $g_{z_{jt}^x} = 0$ for all j and t. Then the ISTC term in (7) vanishes and the BGP growth rate simplifies to:*

$$g_w = g_v = g_{k_n} = \frac{(\Psi g_z)_n}{(\Psi\Gamma(I-\alpha)\mathbf{1})_n}. \tag{34}$$

The key equilibrium growth rates simplify to:

$$g_{p_j^x} = 0 \qquad \forall\, j, \tag{35}$$

$$g_{p_j} = -(\Psi g_z)_j + (\Psi\Gamma(I-\alpha)\mathbf{1})_j\, g_w \qquad \forall\, j, \tag{36}$$

$$g_{R_j} = 0 \qquad \forall\, j; \tag{37}$$

and the conditions for convergence to the numeraire sector reduce to:

$$\textit{Gross substitutes:} \quad \frac{(\Psi g_z)_n}{(\Psi\Gamma(I-\alpha)\mathbf{1})_n} \geq \frac{(\Psi g_z)_j}{(\Psi\Gamma(I-\alpha)\mathbf{1})_j} \qquad \forall\, j \neq n, \tag{38}$$

$$\textit{Gross complements:} \quad \frac{(\Psi g_z)_n}{(\Psi\Gamma(I-\alpha)\mathbf{1})_n} \leq \frac{(\Psi g_z)_j}{(\Psi\Gamma(I-\alpha)\mathbf{1})_j} \qquad \forall\, j \neq n. \tag{39}$$

Proof. Setting $g_{z^x} = \mathbf{0}$ and $g_{z_n^x} = 0$: the ISTC term in (7) vanishes, giving (34). Investment prices satisfy $g_{p_i^x} = g_{z_n^x} - g_{z_j^x} = 0$ for all i, giving (35), and rental rates follow as $g_{R_j} = 0$ for all j, giving (37). The numeraire condition pins down $g_{p_n} = 0$, so both the investment price and the output price of sector n are constant. Sectoral prices follow from (9) with $a = -\Psi g_z$ and $b = \Psi\Gamma(I-\alpha)\mathbf{1}$, giving (36). The consistency conditions (SC-GS) and (SC-GC) become $(0-a_j)/b_j \lesseqgtr (0-a_n)/b_n$, i.e., $(\Psi g_z)_j/(\Psi\Gamma(I-\alpha)\mathbf{1})_j \lesseqgtr (\Psi g_z)_n/(\Psi\Gamma(I-\alpha)\mathbf{1})_n$, giving (38) and (39) respectively. □

A.2 Equilibrium Outcomes, Open Economy

Proof Proposition (open ec) 2. Use the optimality conditions of the firm, and rewrite the expenses in different intermediate and investment goods as a function of gross output, i.e.,

$$\mu_{ni}(1-\gamma_i)p_{it}y_{it} = p_{nt}m_{nit}$$

$$\alpha_i\gamma_i p_{it}y_{it} = R_{it}k_{it}$$

$$(1-\phi_{jt})\omega_{ji}p_{it}^x x_{it} = p_{jt}\chi_{jit}^d$$

Under no arbitrage, the user cost of capital satisfies

$$R_{it} = p_{it-1}^x\left[\frac{1}{R_t} - (1-\hat{\delta}_i)\frac{p_{it}^x}{p_{it-1}^x}\right]$$

where $1-\hat{\delta}_i$ corresponds to the adjusted undepreciated value of a unit of capital adjusted along the BGP, i.e., $1-\hat{\delta}_i \equiv \frac{1-\delta_i}{1+g_{k_i}}$, and $R_t = \beta\frac{U'(c_t)}{U'(c_{t-1})}$ is the interest rate in the economy.

Combining the optimality conditions for capital and investment, as well as the steady-state level of capital

$$\alpha_i\gamma_i p_{it}y_{it} = \left[\frac{1}{R_t} - (1-\hat{\delta}_i)\frac{p_{it}^x}{p_{it-1}^x}\right]\frac{p_{jt-1}\chi_{jit-1}^d}{(1-\phi_{jt})\omega_{ji}}\frac{x_{it}}{x_{it-1}}\frac{k_{it}}{x_{it}},$$

which we can use to write the feasibility constraint in each sector n,

$$p_{nt}y_{nt} = p_{nt}c_{nt} + \sum_i p_{nt}\chi_{nit}^d + \sum_j p_{nt}m_{njt}.$$

Then,

$$\tilde{\zeta}_{nt}\frac{y_{nt}}{c_{nt}} = \tilde{\zeta}_{nt} + \sum_i \frac{\alpha_i\gamma_i(1-\phi_{nt})\omega_{ni}}{\frac{1}{R_t} - (1-\hat{\delta}_i)\frac{p_{it+1}^x}{p_{it}^x}}\frac{x_{it+1}}{k_{it+1}}\frac{x_{it}}{x_{it+1}}\frac{p_{it+1}y_{it+1}}{p_{it}y_{it}}\tilde{\zeta}_{it}\frac{y_{it}}{c_{it}} + \sum_j(1-\gamma_j)\mu_{njt}\tilde{\zeta}_{jt}\frac{y_{jt}}{c_{jt}}.$$

This is a system of equations across sectors that can be solved for the Domar weights $\tilde{\eta}_n \equiv \tilde{\zeta}_n \frac{y_n}{c_n}$.

Along the BGP, the solution satisfies

$$\left[I - \tilde{\beta}^{-1}\hat{\delta}\boldsymbol{\Gamma}\boldsymbol{\alpha}(I-\boldsymbol{\phi})\boldsymbol{\Omega} - (I-\boldsymbol{\Gamma})\boldsymbol{M}\right]^{-1}\tilde{\boldsymbol{\zeta}} \equiv \tilde{\boldsymbol{\eta}} \tag{40}$$

where $\tilde{\beta}$ is a vector of effective discount factors, with typical element $\tilde{\beta}_i \equiv \frac{1}{\beta} - (1-\hat{\delta}_i)$, and the vector of depreciation rates contains typical element $\hat{\delta}_i$. □

Proof Proposition (open ec) 3. The planner's problem associated with our economy is

$$W \equiv \max_{C_t, Y_t, \omega_{int}, \chi_{int}, x_{nt}, k_{nt+1}, m_{int}, \epsilon_t, \epsilon_t^f} \sum_{t=0}^{\infty} \beta^t \ln(C_t)$$

In this proof, $\tilde{z}_{nt} \equiv \exp(z_{nt})$ denotes level TFP in sector n. Subject to

$$y_{nt} = \left(\frac{\tilde{z}_{nt}\left(\frac{k_{nt}}{\alpha_n}\right)^{\alpha_n}\left(\frac{l_{nt}}{1-\alpha_n}\right)^{1-\alpha_n}}{\gamma_n}\right)^{\gamma_n}\left(\frac{m_{nt}}{1-\gamma_n}\right)^{1-\gamma_n}, \qquad \text{for } \gamma_n \in [0,1],$$

$$k_{nt+1} = x_{nt} + (1-\delta_n)k_{nt},$$

$$x_{nt} = \prod_{i=1}^{N}\left(\frac{\chi_{int}}{\omega_{int}}\right)^{\omega_{int}}, \quad \sum_{in}\xi_{in}\omega_{int}^{\iota_n} = B_n,$$

$$y_{nt} = c_{nt} + \sum_i m_{nit} + \sum_i \chi_{nit}^d,$$

$$Y_t = \prod_{n=1}^{N}\left(\frac{c_{nt}}{\theta_n}\right)^{\theta_n}, \qquad \sum_{n=1}^{N}\theta_n = 1 \text{ and } \quad \theta_n > 0;$$

$$Y_t = C_t + \epsilon_t, \qquad \epsilon_t - \frac{\epsilon_t^f}{\tau} = 0$$

$$\epsilon_t^f = \prod_{i=1}^{N}\left(\frac{\chi_{it}^f}{\phi_i^f}\right)^{\phi_i^f} \qquad \chi_{it}^f = \sum_n \chi_{int}^f,$$

$$\chi_{int} = \left(\frac{\chi_{int}^d}{1-\phi_i}\right)^{1-\phi_i}\left(\frac{\chi_{int}^f}{\phi_i}\right)^{\phi_i}.$$

where we have defined $\tilde{z} \equiv \exp z$ for notational convenience.

The envelope condition then yields that

$$\frac{\partial C}{\partial \tilde{z}_{nt}}\tilde{z}_{nt} = \lambda_{nt} y_{nt}\frac{\partial y_{nt}}{\partial \tilde{z}_{nt}}\frac{\tilde{z}_{nt}}{y_{nt}}$$

where λ_n is the Lagrange multiplier associated with the feasibility constraint for good n and the last term in the above equation is simply the elasticity of gross output to productivity, i.e., γ_n. We can rewrite this in terms of the change in welfare, which is proportional to $d\ln(C_t)$ because utility is separable in time.

$$\frac{\partial C_t}{\partial \tilde{z}_{nt}} \frac{\tilde{z}_{nt}}{C_t} = \frac{\nu_t}{C_t} \frac{\lambda_{nt} y_{nt}}{\nu_t} \gamma_n$$

Along the BGP, aggregate consumption is a constant fraction of GDP, ν, and by definition, $\tilde{\eta}_n = \frac{\lambda_{nt} y_{nt}}{\nu_t}$, i.e., the Domar weight.

$$\frac{d\ln C_t}{d\ln \tilde{z}_{nt}} = \frac{\nu_t}{C_t} \tilde{\eta}_{nt} \gamma_n.$$

□

Proof Proposition 4. Use the solution and the definition of $\tilde{\zeta}_i$ to solve for relative prices given investment rates.

$$\frac{p_i}{p_j} = \frac{c_j}{c_i} \frac{\tilde{\zeta}_i}{\tilde{\zeta}_j} = \frac{\tilde{\eta}_i}{\tilde{\eta}_j} \frac{y_j}{y_i}$$

These relative prices are useful to define the demand for intermediate inputs, investment and labor as a function of the vector of sectoral gross output. The demand for intermediate inputs follows $(1-\gamma_i)\frac{\tilde{\eta}_i}{\tilde{\eta}_n} y_n = m_{ni}$, while the demand for domestic investment goods is $\tilde{\beta}_i \frac{x_i}{k_i}(1-\phi_j)\omega_{ji}\alpha_i\gamma_i \frac{\tilde{\eta}_i}{\tilde{\eta}_j} y_j = \chi_{ji}$. The demand for imported investment satisfies

$$\tilde{\beta}_i \frac{x_i}{k_i}(\phi_j)\omega_{ji}\alpha_i\gamma_i \frac{\tilde{\eta}_i}{p^f}\nu = \chi^f_{ji}.$$

Total investment in sector i defines the level of the stock of capital as

$$x_i = \prod_j \left(\tilde{\beta}_i \left(\frac{x_i}{k_i}\alpha_i\gamma_i \frac{\tilde{\eta}_i}{\tilde{\eta}_j} y_j \right)^{1-\phi_j} \left(\frac{x_i}{k_i}\alpha_i\gamma_i \frac{\tilde{\eta}_i}{p^f}\nu \right)^{\phi_j} \right)^{\omega_{ji}},$$

, or equivalently, $k_i = \prod_j \left(\tilde{\beta}_i \left(\alpha_i\gamma_i \frac{\tilde{\eta}_i}{\tilde{\eta}_j} y_j \right)^{1-\phi_j} \left(\alpha_i\gamma_i \frac{\tilde{\eta}_i}{p^f}\nu \right)^{\phi_j} \right)^{\omega_{ji}}$.

Assume that the supply of labor is inelastic at 1, so the fraction of labor allocated to each sector follows Domar weights adjusted by the sectoral labor expenditure shares in gross output,

$$l_i^\star = \frac{(1-\alpha_i)\gamma_i p_i y_i}{\sum_i (1-\alpha_i)\gamma_i p_i y_i} = \frac{(1-\alpha_i)\gamma_i \tilde{\eta}_i}{\sum_i (1-\alpha_i)\gamma_i \tilde{\eta}_i}.$$

For the purpose of describing final demand, it would be useful to define $\tilde{l}_i = \frac{l_i^\star}{\gamma_i(1-\alpha_i)}$.

Final output in each sector is then

$$y_n = \left[\exp(z_n)(\prod_i \left(\tilde{\beta}_i(\frac{\tilde{\eta}_n}{\tilde{\eta}_i}y_i)^{1-\phi_i}(\frac{\tilde{\eta}_n}{p^f}\nu)^{\phi_i}\right)^{\omega_{in}})^{\alpha_n}(\tilde{l}_n)^{1-\alpha_n}\right]^{\gamma_n}\left[\prod_i\left(\frac{\tilde{\eta}_n}{\tilde{\eta}_i}y_i\right)^{\mu_{in}}\right]^{1-\gamma_n}$$

Taking logs and writing output in matrix form, we obtain

$$\ln(\mathbf{y}) = \Gamma z + \boldsymbol{\iota} + \Gamma\alpha\boldsymbol{\phi}'\boldsymbol{\Omega}'\ln(\nu) + \Gamma\alpha(I-\boldsymbol{\phi})'\Omega'\ln(\mathbf{y}) + (I-\Gamma)M'\ln(\mathbf{y})$$

where each element of the vector $\boldsymbol{\iota}$ can be described as $\iota_n \equiv \gamma_n(1-\alpha_n)\ln(\tilde{l}_n) + \gamma_n\alpha_n\sum_i(1-\phi_i)\omega_{in}\ln(\frac{\tilde{\eta}_n}{\tilde{\eta}_i}) + \gamma_n\alpha_n\sum_i\phi_i\omega_{in}\ln(\frac{\tilde{\eta}_n}{p^f}) + \gamma_n\alpha_n\sum_i\omega_{in}\ln(\tilde{\beta}_i) + (1-\gamma_n)\sum_i\mu_{in}\ln(\frac{\tilde{\eta}_n}{\tilde{\eta}_i})$.

The solution for gross output is then

$$\ln(\mathbf{y}) = \Xi\Gamma z + \Xi\boldsymbol{\iota} + \Xi\Gamma\alpha\boldsymbol{\phi}'\boldsymbol{\Omega}'\ln(\nu) \tag{41}$$

where the multiplier on sectoral productivity is $\Xi \equiv (I - \Gamma\alpha(I-\boldsymbol{\phi})'\Omega' - (I-\Gamma)M')^{-1}$. Let the price level of the economy be normalized to $p = 1$; then, aggregate value added is $\nu = \frac{p_n y_n}{\tilde{\eta}_n}$ for any n. We can compute a geometric average of each of the terms using the expenditure shares of consumption and investment $\zeta_n \equiv \frac{p_n\hat{c}_n}{\nu} + \frac{p_n x_n^d}{\nu}$ as weights. Note that $p_n\hat{c}_n$ is the value of final uses from sector n that are allocated to aggregate consumption (these values can be split due to the CRS aggregator for final uses). Hence, weights sum to 1 since trade is balanced.

$$\ln(\nu) = \sum_n \zeta_n\ln(p_n) + \sum_n\zeta_n\ln(y_n) - \sum_n\zeta_n\ln(\tilde{\eta}_n).$$

Given a CRS aggregator of sectoral output, the price index for final goods satisfies $\ln(p) = \sum_n\zeta_n\ln(p_n)$. Because final output is the numeraire, the log of the price index equals zero, and therefore, the first term in the expression for value added drops out. The weighting of the terms in the sum also includes investment shares in value added. Investment shares are proportional to consumption shares in value added whenever sectoral value-added shares are proportional to consumption shares across sectors. This is by construction the assumption in canonical models of input–output linkages without capital, and we assume that feature here.[26]

[26] Alternatively, one can set up the economy so that investment in different capital types is produced through the final good. This economy would also allow us to define the price of value added as a function of sectoral prices in a way that they drop out from the expression above while allowing for investment shares that need not be proportional to consumption shares. The undesirable feature of this economy is that sectors producing for final production and intermediate inputs are decoupled from those producing investment.

We have characterized the solution to the last two terms in Equations 40 and 41.

$$\ln(\nu) = \boldsymbol{\zeta}'\Xi(\Gamma \boldsymbol{z} + \boldsymbol{\iota} + \Gamma\alpha\boldsymbol{\phi}'\boldsymbol{\Omega}'\ln(\nu)) - \sum_n \zeta_n \ln(\tilde{\eta}_n) \tag{42}$$

where we can define the elasticity of value to sectoral TFP as $\eta \equiv \boldsymbol{\zeta}'\Xi$. Unlike the Domar weight, these elasticities are not adjusted by the investment rate.

Because of the presence of tradable investment goods we obtain an additional amplification (as in Jones (2011) for tradable intermediate inputs). The reason is that as productivity increases within the economy, export capacity improves, and due to trade balance, this implies higher investment imports. The stronger the dependence on imported equipment and the intensity of capital use, the stronger this amplification channel is.

$$\ln(\nu) = \left(I - \boldsymbol{\zeta}'\Xi\Gamma\alpha\boldsymbol{\phi}'\boldsymbol{\Omega}'\right)^{-1}\left[\boldsymbol{\zeta}'\Xi(\Gamma\boldsymbol{z} + \boldsymbol{\iota}) - \sum_n \zeta_n \ln(\tilde{\eta}_n)\right] \tag{43}$$

Unpacking the vectors, $\zeta_n = \eta_n - \sum_j \gamma_j\alpha_j(1-\phi_j)\omega_{nj}\eta_j - \sum_j(1-\gamma_j)\mu_{nj}\eta_j$

$$\sum_n \zeta_n \ln(\mu_n) = \sum_n \eta_n \ln(\tilde{\eta}_n) - \sum_n\sum_j \gamma_n\alpha_n(1-\phi_n)\omega_{nj}\eta_j \ln(\mu_n) - \sum_n\sum_j(1-\gamma_n)\mu_{nj}\eta_j\ln(\mu_n)$$

Now consider the term $\eta\iota$

$$\begin{aligned}\sum_n \eta_n\iota_n \;=\; & \sum_n(\eta_n\gamma_n(1-\alpha_n)\ln(\tilde{l_n}) + \eta_n\gamma_n\alpha_n\sum_j(1-\phi_j)\omega_{jn}\ln(\frac{\tilde{\eta}_n}{\tilde{\eta}_j}) + \gamma_n\alpha_n\sum_j\phi_j\omega_{jn}\ln(\frac{\tilde{\eta}_n}{p^f}) \\ & + \eta_n(1-\gamma_n)\sum_j\mu_{jn}\ln(\frac{\tilde{\eta}_n}{\tilde{\eta}_j})) + \gamma_n\alpha_n\sum_i\omega_{in}\ln(\tilde{\beta}_i)\end{aligned}$$

which can be rewritten as

$$\begin{aligned}\sum_n \eta_n\iota_n \;=\; & \sum_n \eta_n\gamma_n(1-\alpha_n)\ln(\tilde{l_n}) + \sum_n\eta_n(\gamma_n\alpha_n + 1 - \gamma_n)\ln(\tilde{\eta}_n) + \gamma_n\alpha_n\sum_i\omega_{in}\ln(\tilde{\beta}_i) \\ & - \sum_n\eta_n\gamma_n\alpha_n\sum_j\omega_{jn}\left((1-\phi_j)\ln(\tilde{\eta}_j) + \phi_j\ln(p^f)\right) - \sum_n\eta_n(1-\gamma_n)\sum_j\mu_{jn}\ln(\tilde{\eta}_j)\end{aligned}$$

Therefore, the difference in the last two terms in the expression for value added are

$$\begin{aligned}\sum_n \eta_n\iota_n - \sum_n\zeta_n\ln(\tilde{\eta}_n) \;=\; & \sum_n\eta_n\gamma_n(1-\alpha_n)(\ln(\tilde{l_n}) - \ln(\tilde{\eta}_n)) - \sum_n\eta_n\gamma_n\alpha_n\sum_j\omega_{jn}\phi_j\ln(p^f) \\ & + \gamma_n\alpha_n\sum_i\omega_{in}\ln(\tilde{\beta}_i)\end{aligned}$$

The second to last term can be written as a function of the terms of trade for imported equip-

ment j, $\ln(\tau_j) = \ln(p) - \ln(p^f)$. Because the final good is the numeraire, p=1. Hence,

$$\begin{aligned}\sum_n \eta_n \iota_n - \sum_n \zeta_n \ln(\tilde{\eta}_n) &= \sum_n \eta_n \gamma_n (1-\alpha_n)(\ln(\tilde{I}_n) - \ln(\tilde{\eta}_n)) + \sum_n \eta_n \gamma_n \alpha_n \sum_j \omega_{jn} \phi_j \ln(\tau_j) \\ &\quad + \gamma_n \alpha_n \sum_i \omega_{in} \ln(\tilde{\beta}_i)\end{aligned}$$

which proves our result. □

B Data Appendix

B.1 Data Sources and Coverage

Table 4: Investment Network: Data Sources

Data description	Source
Sector–equipment bridge	US Bridge Tables (BEA)
Number of tools per worker by occupation	Caunedo *et al.* (2023a)
Investment production by sector	WIOD; OECD; Mensah and de Vries (2023)
Input–output structure	WIOD; OECD; Mensah and de Vries (2023)
Employment by occupation and sector	IPUMS; ILOSTAT; PIAAC

Notes: Data sources to construct investment networks. WIOD = World Input–Output Database; OECD = Organisation for Economic Co-operation and Development; BEA = U.S. Bureau of Economic Analysis; IPUMS = Integrated Public Use Microdata Series; ILOSTAT = ILO Statistics; PIAAC = Programme for the International Assessment of Adult Competencies. "Investment production by sector" and "Input–output structure" come from WIOD, OECD and national Supply–Use / IO tables (Mensah and de Vries (2023)). The sector–commodity bridge uses BEA bridge tables to map commodities to industries. "Tools per worker" by occupation is from Caunedo *et al.* (2023a). Employment by occupation and sector combines household microdata (IPUMS), labor statistics (ILOSTAT), and skills microdata (PIAAC).

Table 5: Country Sample and Data Sources

	Country	GDP per capita (2005 PPP)	Use/IO Source	Employment Source	Investment Network Available Years
1	Ethiopia	679	MDV	ILOSTAT	1990–2019
2	Rwanda	1,246	MDV	ILOSTAT	1990–2019
3	Tanzania	1,507	MDV	ILOSTAT	1990–2019
4	Zambia	1,710	MDV	IPUMS	1990–2019
5	Kenya	1,972	MDV	ILOSTAT	1990–2019
6	Cambodia	2,048	OECD	ILOSTAT	2005–2015
7	Senegal	2,728	MDV	IPUMS	1990–2019
8	India	2,872	WIOD	IPUMS	1965–2000; 2000–2014
9	Vietnam	3,128	OECD	IPUMS	2005–2015
10	Ghana	3,219	MDV	ILOSTAT	1990–2019
11	Nigeria	3,481	MDV	IPUMS	1990–2019
12	Philippines	4,366	OECD	IPUMS	2005–2015
13	Indonesia	4,602	WIOD	IPUMS	2000–2014
14	Morocco	4,672	OECD	IPUMS	2005–2015
15	China	6,681	WIOD	IPUMS	1965–2000; 2000–2014
16	Peru	6,832	OECD	PIAAC	2005–2015
17	Colombia	8,367	OECD	ILOSTAT	2005–2015
18	Tunisia	9,353	OECD	ILOSTAT	2005–2015
19	Brazil	9,610	WIOD	IPUMS	1965–2000; 2000–2014
20	Thailand	10,293	OECD	IPUMS	2005–2015
21	South Africa	11,311	OECD	IPUMS	2005–2015
22	Costa Rica	11,580	OECD	IPUMS	2005–2015
23	Turkey	13,941	WIOD	PIAAC	2000–2014
24	Argentina	14,247	OECD	ILOSTAT	2005–2015
25	Mauritius	14,325	MDV	IPUMS	1990–2019
26	Chile	14,534	OECD	PIAAC	2005–2015
27	Mexico	15,230	WIOD	PIAAC	1965–2000; 2000–2014
28	Russia	15,450	WIOD	PIAAC	2000–2014
29	Poland	16,838	WIOD	PIAAC	2000–2014
30	Malaysia	17,412	OECD	IPUMS	2005–2015
31	Lithuania	17,646	WIOD	ILOSTAT	2000–2014
32	Slovakia	20,168	OECD	PIAAC	2000–2014
33	Hungary	20,819	WIOD	PIAAC	2000–2014
34	Czechia	26,624	WIOD	PIAAC	2000–2014
35	Portugal	27,149	WIOD	IPUMS	1965–2000; 2000–2014
36	Slovenia	28,821	OECD	PIAAC	2000–2014
37	Greece	30,138	WIOD	PIAAC	1965–2000; 2000–2014
38	South Korea	30,784	WIOD	PIAAC	1965–2000; 2000–2014
39	New Zealand	31,485	OECD	PIAAC	2005–2015
40	Israel	32,358	OECD	PIAAC	2005–2015
41	Spain	32,769	WIOD	PIAAC	1965–2000; 2000–2014
42	Cyprus	33,025	OECD	ILOSTAT	2005–2015
43	Italy	36,167	WIOD	PIAAC	1965–2000; 2000–2014
44	France	36,651	WIOD	PIAAC	1965–2000; 2000–2014
45	Japan	38,466	WIOD	PIAAC	1965–2000; 2000–2014
46	Germany	38,475	WIOD	PIAAC	1965–2000; 2000–2014
47	Belgium	39,220	WIOD	PIAAC	1965–2000; 2000–2014
48	United Kingdom	39,308	WIOD	PIAAC	1965–2000; 2000–2014
49	Denmark	40,344	WIOD	PIAAC	1965–2000; 2000–2014
50	Sweden	40,381	WIOD	PIAAC	1965–2000; 2000–2014
51	Austria	41,678	WIOD	ILOSTAT	1965–2000; 2000–2014
52	Australia	43,333	WIOD	ILOSTAT	1965–2000; 2000–2014
53	Netherlands	44,662	WIOD	PIAAC	1965–2000; 2000–2014
54	Ireland	47,211	WIOD	PIAAC	1965–2000; 2000–2014
55	Switzerland	49,859	WIOD	IPUMS	2000–2014
56	Norway	54,200	WIOD	PIAAC	2000–2014
57	United States	54,210	WIOD	PIAAC	1965–2000; 2000–2014
58	Singapore	63,949	OECD	PIAAC	2005–2015

Notes: Country-year coverage and main sources of data. "Use/IO Source" refers to VA shares and GFCF imported shares. GDP per capita in chained PPPs (2017 international dollars), 2005.

Table 6: Aggregate Sectors Definition

ISIC Rev.4 Industry Code	Industry Description	Aggregate Sector
A01	Crop and animal production, hunting and related service activities	Agriculture
A02	Forestry and logging	Agriculture
A03	Fishing and aquaculture	Agriculture
B	Mining and quarrying	Mining
C10–C12	Manufacture of food products, beverages and tobacco products	Non-Durables
C13–C15	Manufacture of textiles, wearing apparel and leather products	Non-Durables
C16	Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials	Durables
C17	Manufacture of paper and paper products	Non-Durables
C18	Printing and reproduction of recorded media	Non-Durables
C19	Manufacture of coke and refined petroleum products	Non-Durables
C20	Manufacture of chemicals and chemical products	Non-Durables
C21	Manufacture of basic pharmaceutical products and pharmaceutical preparations	Non-Durables
C22	Manufacture of rubber and plastic products	Non-Durables
C23	Manufacture of other non-metallic mineral products	Durables
C24	Manufacture of basic metals	Durables
C25	Manufacture of fabricated metal products, except machinery and equipment	Durables
C26	Manufacture of computer, electronic and optical products	Electronics
C27	Manufacture of electrical equipment	Electronics
C28	Manufacture of machinery and equipment n.e.c.	Machinery
C29	Manufacture of motor vehicles, trailers and semi-trailers	Transportation
C30	Manufacture of other transport equipment	Transportation
C31, C32	Manufacture of furniture; other manufacturing	Durables
C33	Repair and installation of machinery and equipment	Durables
D35	Electricity, gas, steam and air conditioning supply	Services
E36	Water collection, treatment and supply	Services
E37–E39	Sewerage; waste collection, treatment and disposal activities; materials recovery; remediation activities and other waste management services	Services
F	Construction	Construction
G45	Wholesale and retail trade and repair of motor vehicles and motorcycles	Services
G46	Wholesale trade, except of motor vehicles and motorcycles	Services
G47	Retail trade, except of motor vehicles and motorcycles	Services
H49	Land transport and transport via pipelines	Trpt Services
H50	Water transport	Trpt Services
H51	Air transport	Trpt Services
H52	Warehousing and support activities for transportation	Trpt Services
H53	Postal and courier activities	Trpt Services
I	Accommodation and food service activities	Services
J58	Publishing activities	ICT
J59, J60	Motion picture, video and television programme production, sound recording and music publishing activities; programming and broadcasting activities	ICT
J61	Telecommunications	ICT
J62, J63	Computer programming, consultancy and related activities; information service activities	ICT
K64	Financial service activities, except insurance and pension funding	Services
K65	Insurance, reinsurance and pension funding, except compulsory social security	Services
K66	Activities auxiliary to financial services and insurance activities	Services
L68	Real estate activities	Services
M69, M70	Legal and accounting activities; activities of head offices; management consultancy activities	ICT
M71	Architectural and engineering activities; technical testing and analysis	ICT
M72	Scientific research and development	ICT
M73	Advertising and market research	ICT
M74, M75	Other professional, scientific and technical activities; veterinary activities	ICT
N	Administrative and support service activities	ICT
O84	Public administration and defence; compulsory social security	Services
P85	Education	Services
Q	Human health and social work activities	Services
R, S	Other service activities	Services
T	Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use	Services
U	Activities of extraterritorial organizations and bodies	Services

Notes: Crosswalk between ISIC Rev.4 classification and our 10-sector disaggregation. "Trpt Services" refers to Transportation Services.

Table 7: Country Groups

Low income	Middle income	High income
Ethiopia	Argentina	Australia
Ghana	Brazil	Austria
India	China	Belgium
Kenya	Colombia	Switzerland
Morocco	Costa Rica	Chile
Nigeria	Indonesia	Cyprus
Philippines	Mexico	Czech Republic
Rwanda	Mauritius	Germany
Senegal	Malaysia	Denmark
Tanzania	Peru	Spain
Zambia	Thailand	France
	Turkey	United Kingdom
	South Africa	Greece
		Hungary
		Ireland
		Israel
		Italy
		Japan
		South Korea
		Lithuania
		Netherlands
		Norway
		New Zealand
		Poland
		Portugal
		Russia
		Singapore
		Slovakia
		Slovenia
		Sweden
		United States

Notes: Countries are grouped into income categories per the World Bank classification. *Low income* comprises "Low income" and "Lower middle income"; *Middle income* corresponds to "Upper middle income"; *High income* corresponds to "High income." Average real GDP per capita (chained PPPs, 2017 international dollars) for 2005, from the Penn World Table (PWT): low 2587; middle 11110; high 35056.

B.2 Data Sources for Income Accounting

To estimate sectoral influence in equation 15 we need data on sectoral value-added shares in gross output ($\mathbf{\Gamma}$), sectoral value-added shares ($\tilde{\boldsymbol{\zeta}}$), capital shares in value added ($\boldsymbol{\alpha}$), and the sectoral imported share of investment ($\boldsymbol{\phi}$), together with estimates of input-output linkages ($\boldsymbol{M}$) and our newly estimated investment networks ($\mathbf{\Omega}$), both already described in Section 3.2.

Value-added shares in production (Γ) and sectoral value-added shares (ζ). We compute sectoral value-added shares in gross output and sectoral value-added shares using the same

data sources as the input-output tables for each country: Mensah and de Vries (2023), WIOD, and OECD.

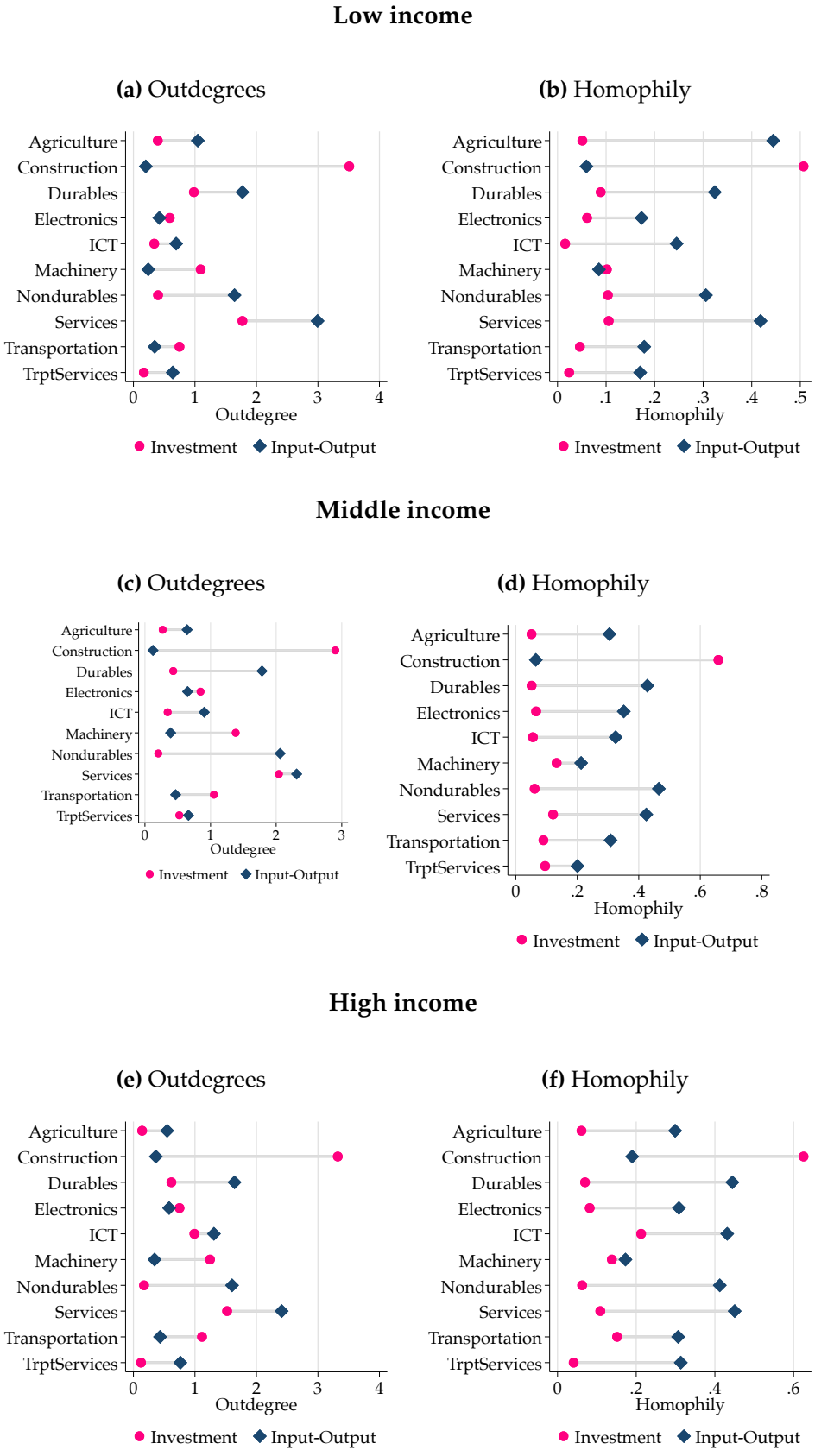
Capital share in value added (α). We exploit data from PWT version 10.01 to compute the labor expenditure share. We estimate capital shares as residuals from labor expenditure shares, under the assumption of CRS value-added production technologies. The capital expenditure share is computed at the aggregate level and therefore country specific but common across sectors.[27]

sectoral imported investment shares (ϕ). For each sector and country, we compute the share of sectoral investment that is sourced from abroad using the *use tables* described in Section 3.2, which contain information on imported and domestic sectoral investment.

[27]Some high- and medium-income countries have data on sectoral capital shares in value added from WIOD. However, we lack this information for more than half of the countries in our sample. For consistency, we use country-specific aggregate capital shares.

B.3 Investmenet Network vs. Input-Output Network

Figure 15: Investment Network vs. Input-Output Network



Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. Circles denote the investment network; diamonds denote the input–output network. Panel (a) reports sectoral outdegrees (row sums); panel (b) reports sectoral homophily (diagonal elements of each network).

Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. Outdegrees are sectoral row sums in the investment network. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Notes: Data for 2005. Concentration index refers to the Herfindahl–Hirschman index computed as the squared sum of the elements of each column of the investment network. The lower the value of the index, the more diversified, with full-diversification achieved at 0.1. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

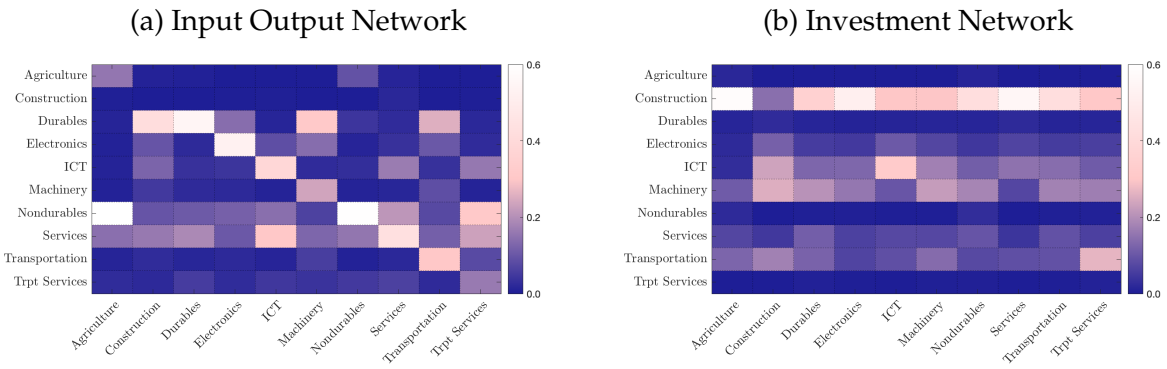
Notes: Data for 2005. Concentration index refers to the Herfindahl–Hirschman index computed as the squared sum of the elements of each column of the input-output network. The lower the value of the index, the more diversified, with full-diversification achieved at 0.1. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. Homophily consists on the diagonal elements of the network, i.e., how important are sectors are provider of investment inputs for themselves. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. Homophily consists on the diagonal elements of the network, i.e., how important are sectors are provider of intermediate inputs for themselves. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. sectoral influence is defined as $\Phi\eta\Gamma$ in Proposition 4, and measures the change in aggregate value added following a change in sectoral productivity. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

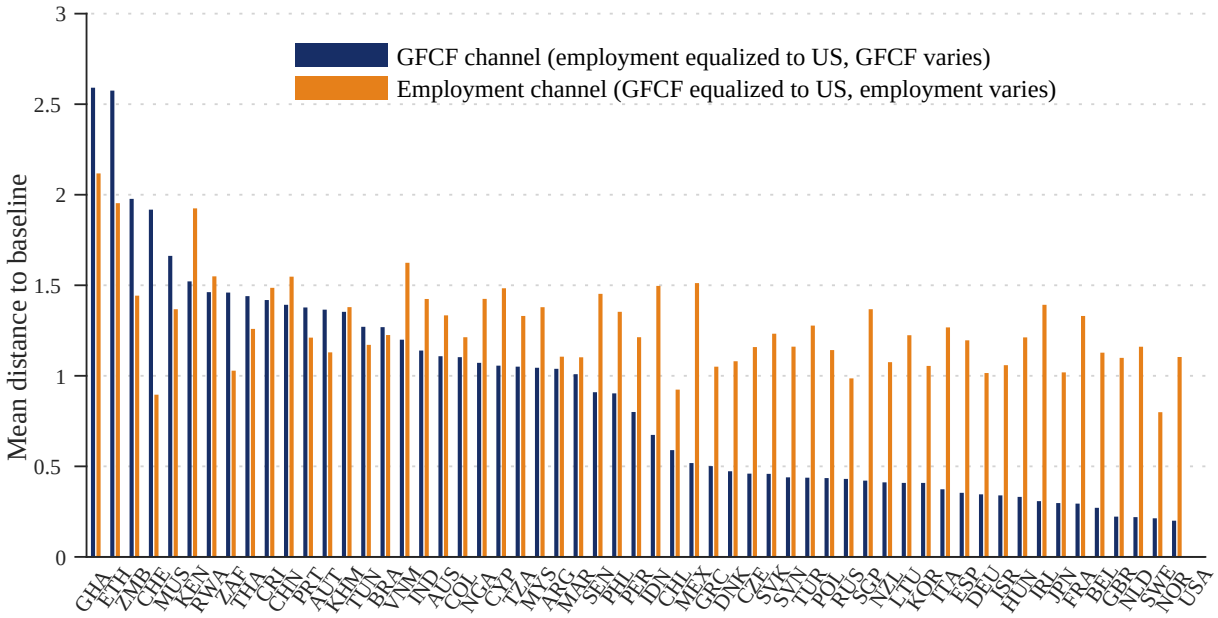
Figure 16: South Korea: Investment Network vs. Input–Output Network, 2014



Notes: Estimates of the investment network and the input–output network in South Korea, 2014.

B.4 Sources of Cross-Country Variation in Investment Networks

Figure 17: Within-country distance between baseline and counterfactual investment networks



Notes: Each bar reports the mean Frobenius distance between the baseline investment network and a counterfactual, averaged over all available years for each country, $\overline{\|\mathbf{\Omega}_t^c - \mathbf{\Omega}_t^{c,\text{CF}}\|_F}$. The navy bars correspond to the GFCF channel: the U.S. occupational employment distribution is imposed on each country while keeping country-own GFCF flows; any deviation from baseline thus reflects GFCF differences. The orange bars correspond to the employment channel: U.S. GFCF levels are imposed while retaining country-own employment composition. Countries are sorted in descending order of their GFCF-channel distance.

B.5 Data Validation

Table 14: Comparison with vom Lehn and Winberry (2022), US 2012

(a) Outdegrees

Sector	This Paper	VLW
Agriculture	0.00	0.00
Construction	1.65	1.31
Durables	0.30	0.24
Electronics	0.90	0.64
ICT	3.02	3.57
Machinery	1.42	1.94
Nondurables	0.04	0.03
Services	1.16	1.06
Transportation	1.35	1.10
Transportation Services	0.16	0.10

(b) Homophily

Sector	This Paper	VLW
Agriculture	0.00	0.00
Construction	0.04	0.03
Durables	0.04	0.05
Electronics	0.12	0.07
ICT	0.43	0.61
Machinery	0.18	0.26
Nondurables	0.01	0.005
Services	0.11	0.12
Transportation	0.11	0.06
Transportation Services	0.03	0.01

Notes: Comparison between investment network estimates in year 2012, corresponding to the year in which we fix the occupational composition of the labor force. This table reports the row sum of the networks, *outdegrees*, as well as the weight of the diagonal in the network, *homophily*.

Table 15: Comparison with Gillmore *et al.* (2025), Chile 2015

	Outdegree		Homophily	
Sector	This Paper	Gillmore *et al.* (2025)	This Paper	Gillmore et al. (2025)
Agriculture	0.32	0.07	0.13	0.04
Construction	4.22	5.13	0.68	0.86
Durables	0.44	0.46	0.07	0.07
Electronics	0.06	0.11	0.01	0.05
ICT	2.73	0.15	0.33	0.03
Machinery	0.49	0.28	0.07	0.11
Nondurables	0.34	0.15	0.09	0.04
Services	1.08	4.47	0.06	0.36
Transportation	0.13	0.04	0.01	0.01
TrptServices	0.20	0.11	0.06	0.04

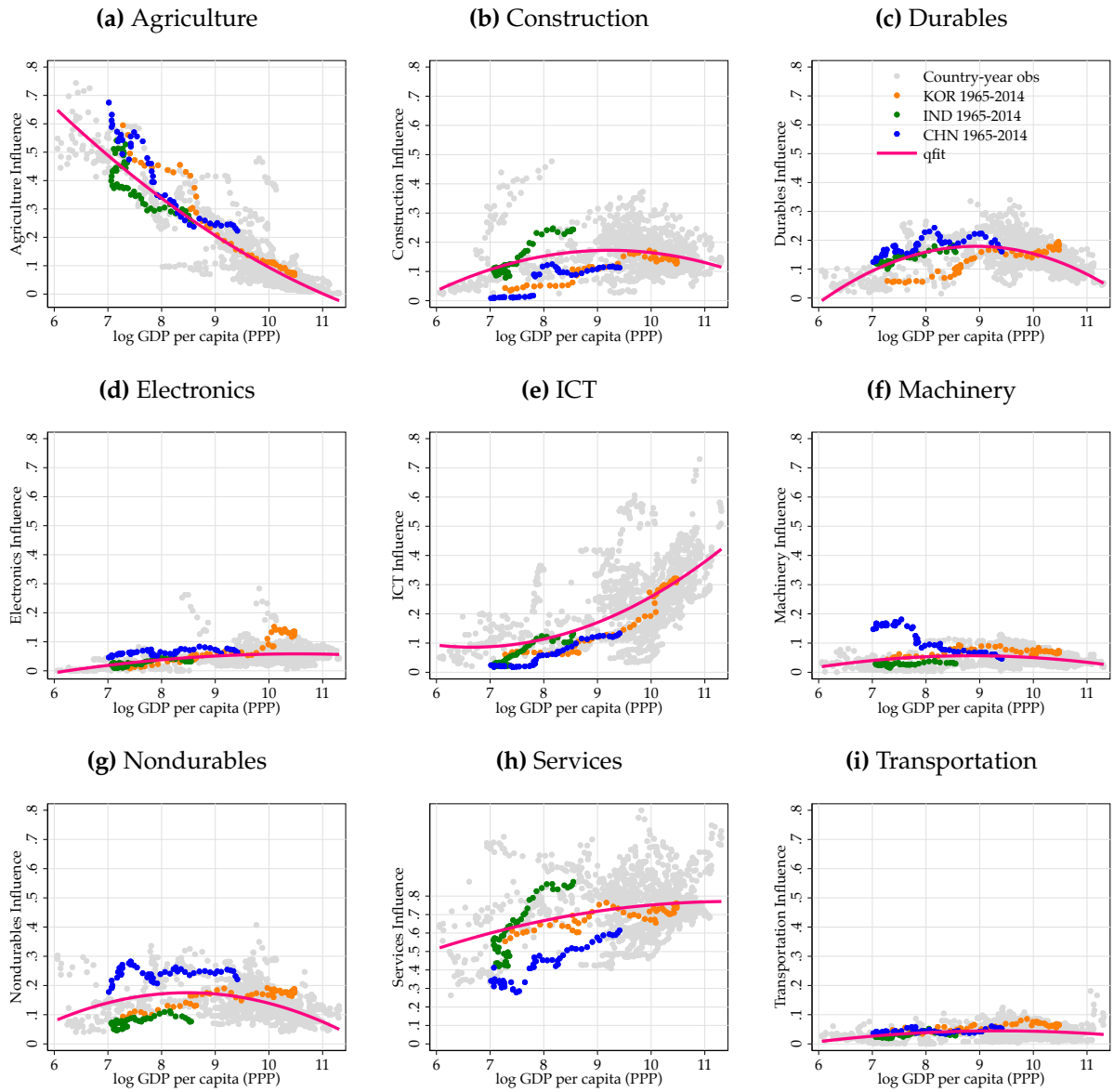
Notes: Comparison between our benchmark (domestic) investment network estimates and Gillmore *et al.* (2025) VAT firm-to-firm transaction estimates. This table reports the row sum of the networks, *outdegrees*, as well as the weight of the diagonal in the network, *homophily*.

B.6 Robustness

We relax the assumption of common cross-country assignment of equipment intensity to occupations by projecting tool usage in each 3-digit occupation to the tasks performed on the job. Then, we exploit cross-country variation in tasks for the same occupation for 19 middle and high income countries, as in Caunedo *et al.* (2023b). We then compute the distance between the network predicted with a common occupation intensity assignment and this cross-country measure. We conclude that a constant assignment explains two thirds of the cross-country variation in investment networks, which suggest that this margin is not driving most of the variation. It is instead the relative flows of investment across sectors and disparities in the 3-digit occupational composition of the labor force.

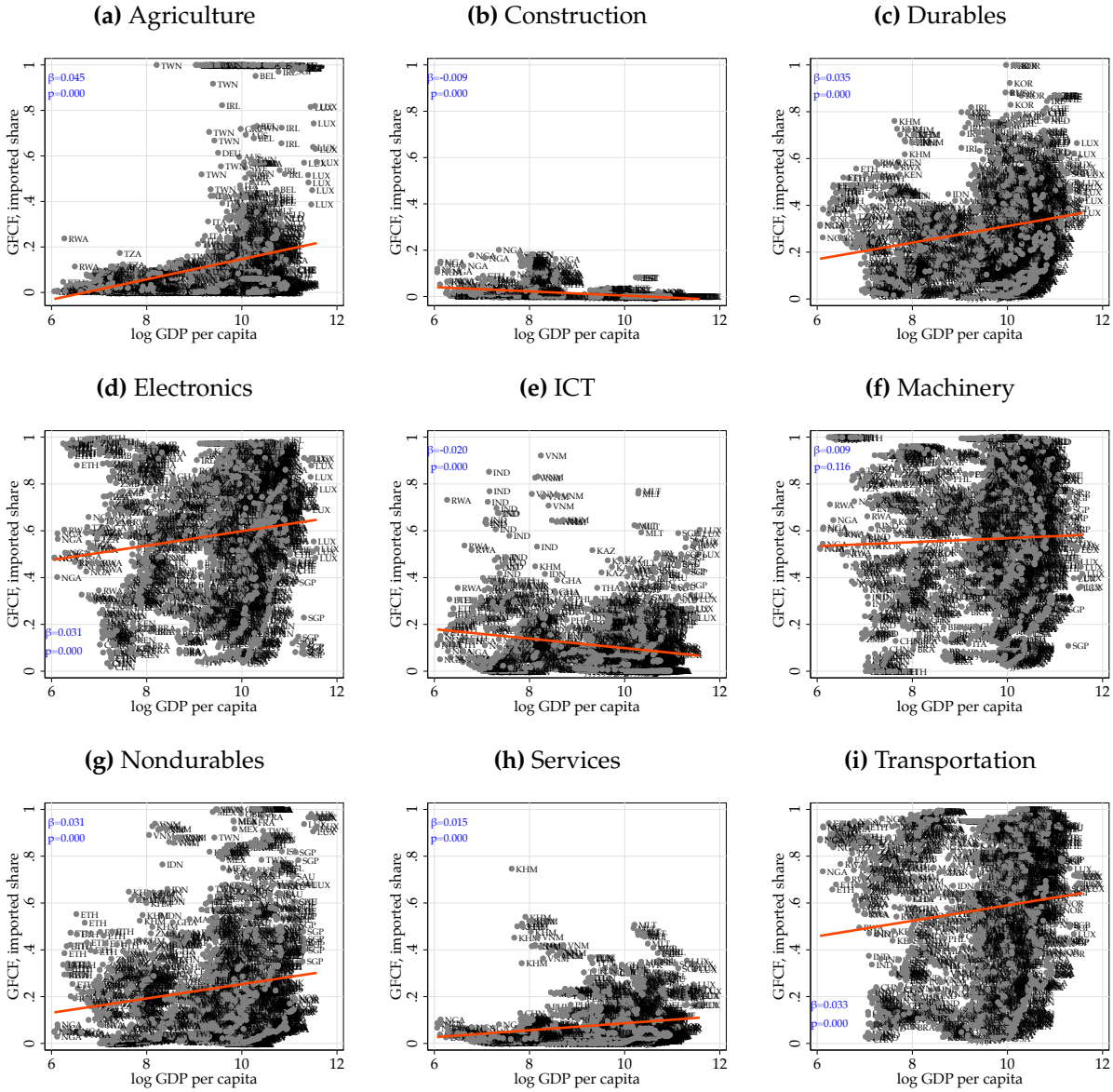
B.7 sectoral Influence Along Development

Figure 18: sectoral Influence Along Development



Notes: Data for all country-years in the sample. sectoral influence is defined as $\Phi\eta\Gamma$ in Proposition 4, and measures the change in aggregate value added following a change in sectoral productivity. Dots represent country–year observations; South Korea (1965–2014), India (1965-2014) and China (1965-2014) are highlighted.

Figure 19: Sectoral Investment, Imported Share



Notes: Data for all country-years in the sample. For each sector, the share of total gross fixed capital formation (GFCF) sourced from abroad. The β coefficient is the slope of the linear fit (in orange) and the p-value is reported under it.

B.8 Transition Dynamics

This appendix reports the time-varying primitives used in the transition calibration of Section 7: the CES investment loadings $\tilde{\omega}_{in,t}$ and the input–output network $\mu_{in,t}$, both reported at the three anchor years 1965, 1990, and 2014. In the transition, $\tilde{\omega}_{in,t}$ is interpolated log-linearly between anchors (column-renormalized at each t) and $\mu_{in,t}$ is linearly interpolated, with both held flat after 2014.

Table 16: Investment Weights, $\tilde{\omega}_{in,t}$ (Korea, 1965/1990/2014)

	1965					1990					2014				
Source i	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.
Agri.	0.44	0.01	0.00	0.39	0.00	0.04	0.00	0.00	0.03	0.00	0.01	0.00	0.00	0.01	0.00
Cons.	0.16	0.13	0.02	0.09	0.61	0.24	0.23	0.07	0.22	0.67	0.32	0.18	0.27	0.52	0.58
ICT	0.00	0.01	0.04	0.01	0.00	0.01	0.03	0.10	0.02	0.01	0.12	0.32	0.42	0.14	0.16
Manuf.	0.17	0.47	0.81	0.30	0.19	0.53	0.51	0.71	0.42	0.24	0.51	0.44	0.24	0.23	0.22
Serv.	0.22	0.38	0.13	0.22	0.20	0.19	0.24	0.12	0.31	0.08	0.04	0.06	0.06	0.10	0.04

Notes: Investment weights recovered to match the data investment-input shares for Korea at each anchor year (closed-form inversion of the CES first-order condition for a common $\iota = 1.20$). Columns are destination sectors and sum to one; rows are source sectors.

Table 17: Input–Output Network, $\mu_{in,t}$ (Korea, 1965/1990/2014)

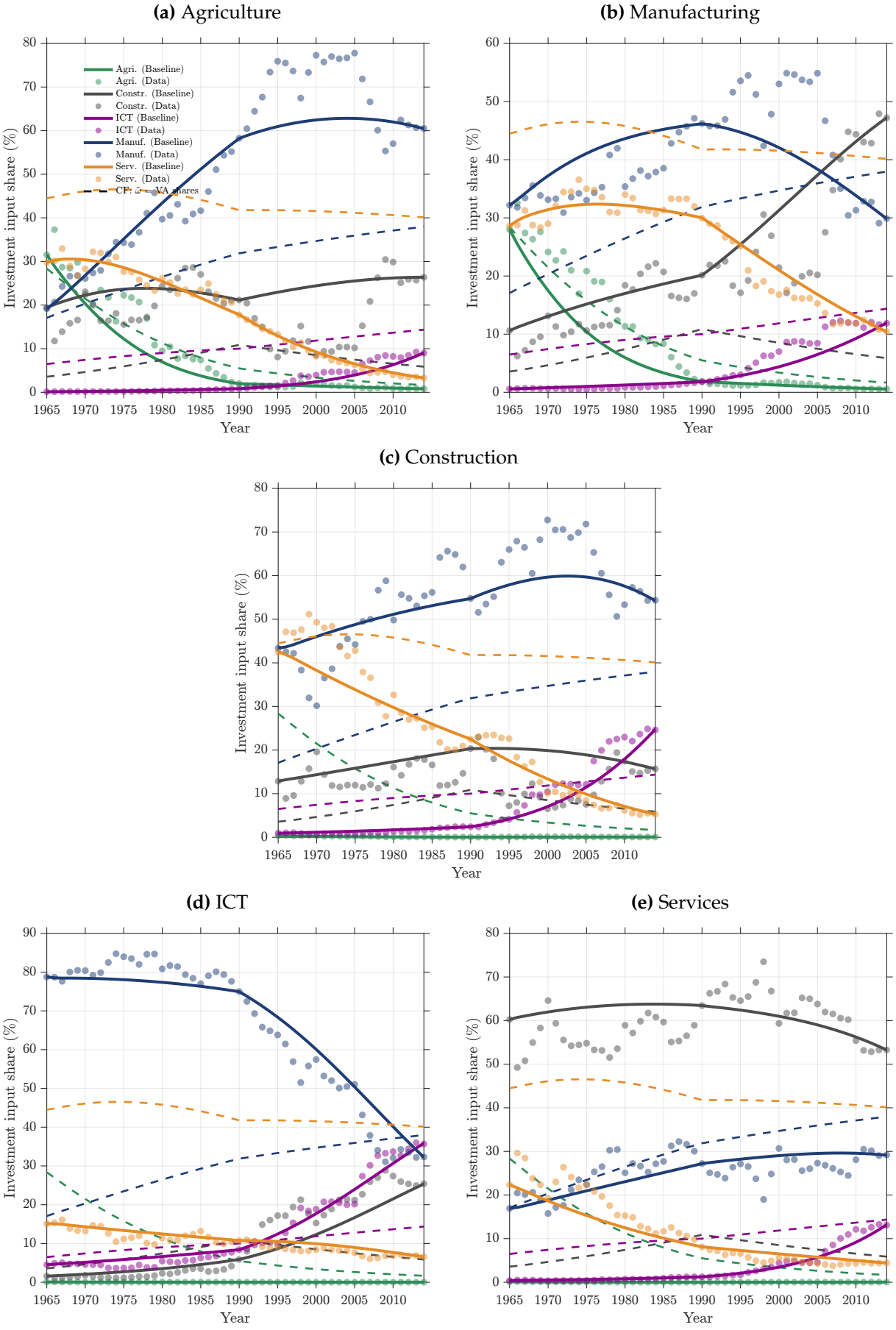
	1965					1990					2014				
Source i	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.
Agri.	0.51	0.01	0.01	0.40	0.01	0.23	0.00	0.00	0.10	0.00	0.16	0.01	0.00	0.03	0.01
Cons.	0.00	0.00	0.00	0.00	0.05	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00	0.02
ICT	0.00	0.03	0.06	0.01	0.07	0.01	0.08	0.16	0.04	0.13	0.01	0.12	0.38	0.03	0.17
Manuf.	0.37	0.60	0.63	0.49	0.24	0.54	0.62	0.58	0.67	0.25	0.66	0.68	0.27	0.75	0.33
Serv.	0.12	0.36	0.31	0.10	0.63	0.22	0.29	0.26	0.19	0.57	0.17	0.18	0.35	0.18	0.48

Notes: 5-sector input–output network for Korea at the three anchor years; data source: KOR WIOD/OECD-consistent IO tables (see Section 3.2). Columns are destination sectors and sum to one.

Counterfactual: Investment Network = VA shares — Implied Investment Input Shares

In the counterfactual where the CES investment loadings $\tilde{\omega}_{in,t}$ are replaced by stacked value-added-share columns, the implied investment input shares for each destination sector mechanically coincide with the value-added share vector. Figure 20 reports the model implied investment input shares by destination sector under this counterfactual against the data.

Figure 20: Counterfactual ($\tilde{\omega}$ = VA shares): Investment Input Shares by Destination Sector

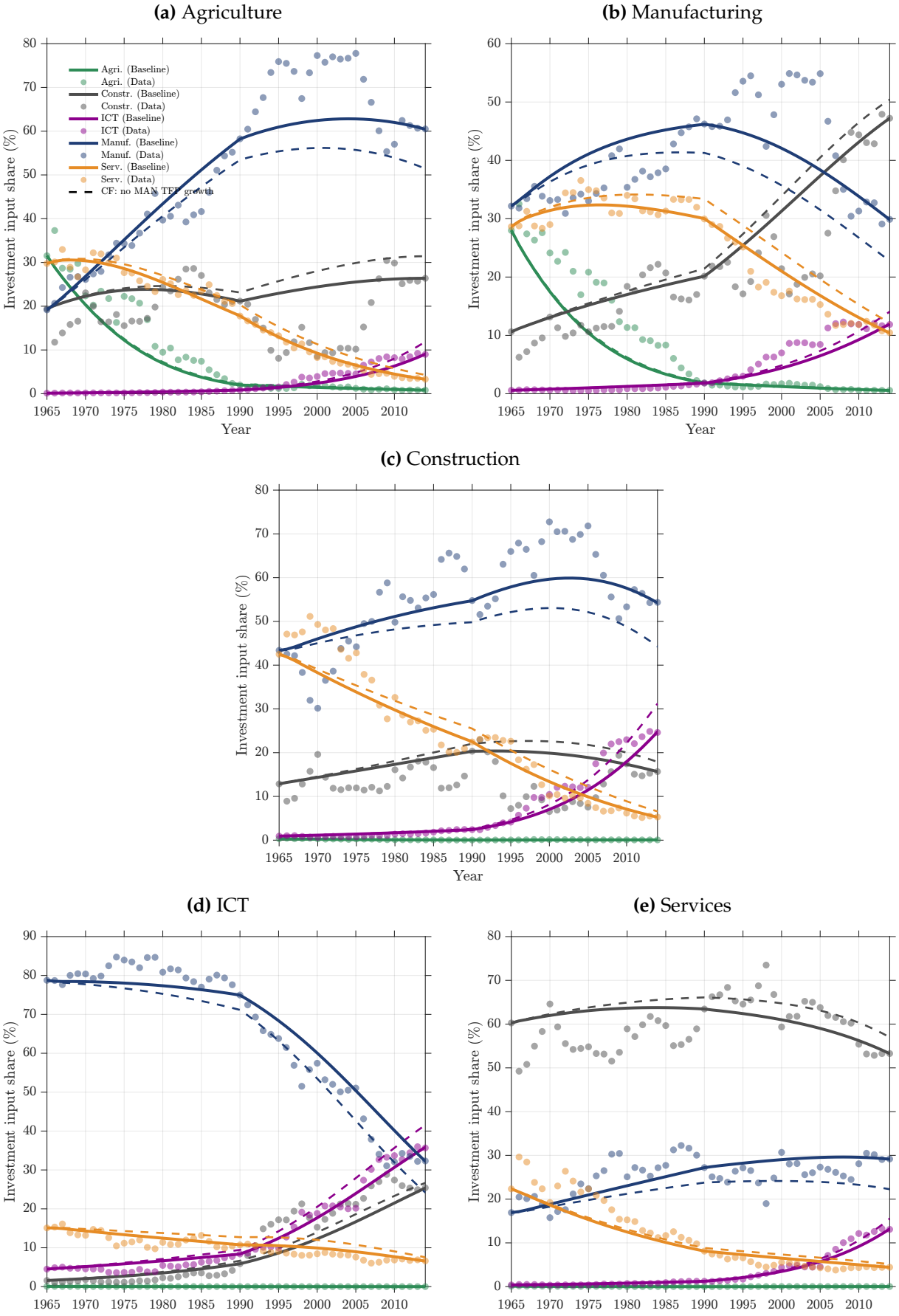


Notes: Solid lines denote the baseline model; dashed lines denote the counterfactual in which $\tilde{\omega}_{in,t}$ is replaced at each anchor by stacked VA-share columns. Markers denote the data.

Counterfactual: No TFP Growth in Manufacturing — Implied Investment Input Shares

In the counterfactual where $z_{\text{MAN},t}$ is held at its 1965 level for the entire transition, the investment loadings $\tilde{\omega}_{in,t}$ are unchanged from the baseline, but the equilibrium investment input shares respond to the resulting relative-price path. Figure 21 reports the model implied investment input shares by destination sector under this counterfactual against the data.

Figure 21: Counterfactual (no MAN TFP growth): Investment Input Shares by Destination Sector



Notes: Solid lines denote the baseline model; dashed lines denote the counterfactual in which $z_{\text{MAN},t}$ is held at its 1965 value for the entire transition. Markers denote the data.